Learning the descriptor, keeping the least squares: a bilevel-trained replacement for the bispectrum in quadratic SNAP

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(code, logs and data pipeline in the accompanying proposal/ directory)

Abstract

Quadratic SNAP expresses the energy of an atom as a quadratic function of hand-built bispectrum components, with coefficients obtained by linear least squares. Its 1595 coefficients at $J_{\max} = 4$ make it accurate when data are plentiful, but it overfits badly on small training sets. We replace the bispectrum by a learned, rotation-, translation- and permutation-invariant descriptor, BLADE, and keep everything else: the quadratic energy form, the isolated-atom constraint, analytic forces and the unweighted least-squares solve. BLADE's 16 channels give a head of 152 coefficients. The descriptor is trained by differentiating through the closed-form least-squares solution, so that it minimises the group-balanced, stratified k-fold cross-validated error of its own head. We use Wood and Thompson's 363-configuration Ta set and their cross-validation protocol (19 fractions, 10 draws each). For each of five training seeds, BLADE's full-data energy error (0.0026–0.0075 eV/atom) is below both the published (0.00898) and the reimplemented (0.0082) value. On the 80/20 split its errors are lower for energies (0.0155 vs. 1.07) and forces (0.0595 vs. 0.0915 eV/Å). Its equal-weight and pooled held-out errors are lower at every fraction, and on the GSF, Liquid and Surface subset it reaches quadratic SNAP's full-data error with 0.55 of the data. Quadratic SNAP keeps the edge in full-data forces and, once overdetermined, on near-crystalline groups. On Mo, Nb and Re, whose datasets are an order of magnitude larger, BLADE is better only at the smallest fractions and on Nb's defects. With abundant data its error stops falling, and it is at parity on Re and behind on Nb and Mo. Invariances hold to round-off on all four metals. Ablations attribute the gains to the out-of-fold objective, group balancing and built-in invariance.

1 Introduction

Machine-learned interatomic potentials of the density-projection family split the map from an atomic environment to an energy into two stages. A descriptor, such as the SNAP bispectrum, ACE or MTP moments, turns the neighbour density of an atom into a fixed-length, symmetry-invariant vector, and a regression model turns that vector into an energy [1–3]. The spectral neighbour analysis potential (SNAP) takes this separation to its logical end: the descriptor is a set of hand-built bispectrum components, and the regression is linear in its coefficients, so the whole fit is one linear least-squares problem [1]. Wood and Thompson extended the regression to second order in the descriptor, the quadratic SNAP form, and showed on tantalum that this lowers the training energy error at $J_{\max} = 4$ from 0.030 to 0.009 eV/atom [4]. The price is coefficients: the 55 bispectrum components now carry 1595 free parameters. In the same paper's cross-validation analysis, held-out errors explode below $\sim 30\%$ of the training data.

In SNAP the descriptor is the only nonlinear, symmetry-carrying ingredient, and it is fixed before any data are seen. This suggests a question that has received little attention. What happens if the

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bispectrum is replaced by a *learned* descriptor while everything that makes SNAP attractive is kept: the per-atom energy form, the exact zero for an isolated atom, analytic forces, and above all the convex least-squares head? A learned descriptor could adapt its few channels to the material and need far fewer head coefficients. It also forfeits guarantees, because invariances and smoothness must now be built into its architecture or learned. Learned representations are normally trained end to end with a nonlinear readout, which makes it impossible to say whether an improvement comes from the representation or from the regression.

We keep the regression *exactly* as in quadratic SNAP and change only the descriptor. Our descriptor, BLADE (Bilevel Least-squares-Adapted DEscriptor), is an invariant, atom-centred density expansion with learned radial channels, power-spectrum and bispectrum contractions, and a small network that compresses them to $K' = 16$ channels. The contribution is how it is trained. The quadratic head is solved in closed form inside the training loop, and the descriptor is updated by back-propagating through that solution so as to minimise the stratified k-fold cross-validated error of its own least-squares head. The descriptor is thus trained for the quantity that matters when data are scarce: how well a least-squares fit on part of the data predicts the rest. At deployment the head is solved by the same FitSNAP call that fits quadratic SNAP, on the same rows, so the comparison isolates the descriptor.

We evaluate the substitution with the cross-validation protocol of the original paper, which was fixed before any result was seen (Sec. 4): 19 training fractions, 10 random draws each, plus the full-data endpoint. Every comparison is paired draw by draw against our FitSNAP reimplementation of quadratic SNAP and anchored to the published value. We repeat the full protocol on three further refractory metals, molybdenum, niobium and rhenium (HCP), taken from an independent curated dataset [5]. We audit rotation, translation and permutation invariance, measure molecular-dynamics cost, and ablate the design choices.

The answer depends on how much data there are relative to quadratic SNAP's 1595 coefficients. When data are scarce, as for the 363 Ta configurations, a descriptor trained for the cross-validated error of its own least-squares head removes quadratic SNAP's overfitting without giving up full-data energy accuracy. When data are abundant, as for the thousands of configurations of each transfer metal, quadratic SNAP is never underdetermined. Its fixed basis with many coefficients then catches up with the learned descriptor, whose error saturates, and on Nb and Mo overtakes it. Built-in invariance, the out-of-fold objective and group balancing are each necessary for the gains. Section 5 reports the results, including where BLADE does *not* improve on quadratic SNAP, and Sec. 6 explains them mechanistically.

2 Quadratic SNAP and the role of its descriptor

Energy form. SNAP writes the potential energy of N atoms as a reference pair potential plus a sum of atomic energies, $E(\{\mathbf{r}\}) = E_{\mathrm{ref}} + \sum_i E^i$, with forces $\mathbf{F}^j = -\nabla_j E$ obtained analytically [1]. In the quadratic form of Wood and Thompson [4],

$$E^i = \boldsymbol{\beta} \cdot \Delta\mathbf{B}^i + \tfrac{1}{2}\,(\Delta\mathbf{B}^i)^{\top} \boldsymbol{\alpha}\, \Delta\mathbf{B}^i, \qquad \Delta\mathbf{B}^i = \mathbf{B}^i - \mathbf{B}^{\varnothing}, \tag{1}$$

where $\mathbf{B}^i \in \mathbb{R}^K$ collects the bispectrum components of the neighbour density of atom i (4D hyperspherical-harmonic expansion of the density within r_{cut}, contracted to rotation invariants), and $\mathbf{B}^{\varnothing}$ is the bispectrum of an isolated atom. The shift by $\mathbf{B}^{\varnothing}$ together with $\beta_0 = 0$ (`bzeroflag=1`) makes the energy of an isolated atom exactly zero. At $J_{\max} = 4$ (LAMMPS `twojmax=8`) there are $K = 55$ components, so Eq. (1) has $K + K(K+1)/2 = 1595$ coefficients. Because E is linear in $(\boldsymbol{\beta}, \boldsymbol{\alpha})$, every per-atom energy row E/N and every force component of every training configuration is a linear equation in the coefficients. The fit is an ordinary least-squares problem $\min \|\mathbf{A}\mathbf{c} - \mathbf{y}\|^2$, where the columns of $\mathbf{A}$ hold configuration sums of $\phi(\Delta\mathbf{B}^i) = [\Delta B^i_k,\ \tfrac{1}{2}(\Delta B^i_k)^2,\ \Delta B^i_k \Delta B^i_l]$ and their negative position derivatives.

The descriptor is the only nonlinear ingredient. Wood and Thompson interpret Eq. (1) as the second-order Taylor expansion of an embedding energy $F(\rho)$ with $\rho = \mathbf{a} \cdot \mathbf{B}$, generalised to a full symmetric $\boldsymbol{\alpha}$ [4]. They also note that it is equivalent to a two-hidden-layer network whose first layer (the bispectrum) is fixed. Everything that is learned is learned linearly; all nonlinearity, body order and symmetry come from $\mathbf{B}$. This has two consequences that the original paper documents and that motivate this work. Accuracy improves steeply with K (roughly $\propto K^{-3}$ in the training error), but the

$\mathcal{O}(K^2)$ coefficients make the fit overdetermined only for large training sets. In their cross-validation (CV) analysis, prediction errors are "extremely large" below $\sim$30% of the 363-configuration Ta training set [4].

The question. If the bispectrum is replaced by a learned map $\mathbf{B}_\theta$ while Eq. (1) and its least-squares solution are kept *exactly*, one can ask three things. Can the learned descriptor match the accuracy of the hand-built one? Can it break the coupling between expressiveness and coefficient count that drives the overfitting? And what does it cost in the symmetries that the bispectrum guarantees by construction?

3 BLADE: a descriptor trained through the least-squares head

BLADE (Bilevel Least-squares-Adapted DEscriptor) replaces $\mathbf{B}^i$ in Eq. (1) by a learned map $\mathbf{B}^i_\theta \in \mathbb{R}^{K'}$. Everything else is unchanged: the per-atom energy form, the isolated-atom shift, the reference potential, the cutoff, the unweighted least-squares problem and the solver. The resulting potential is

$$E = E_{\mathrm{ref}} + \sum_i \Big[\boldsymbol{\beta}\cdot\Delta\mathbf{B}^i_\theta + \tfrac{1}{2}(\Delta\mathbf{B}^i_\theta)^\top \boldsymbol{\alpha}\,\Delta\mathbf{B}^i_\theta\Big], \qquad \Delta\mathbf{B}^i_\theta = \mathbf{B}_\theta(\mathcal{N}_i) - \mathbf{B}_\theta(\varnothing), \tag{2}$$

where $\mathcal{N}_i = \{\mathbf{r}_{ij} : r_{ij} < r_{\mathrm{cut}}\}$ is the neighbourhood of atom i. We call it BLADE-qSNAP.

3.1 Descriptor architecture

Fixed primitive density basis. Each neighbour contributes $g_{n\ell m}(\mathbf{r}_{ij}) = T_n\big(2r_{ij}/r_{\mathrm{cut}} - 1\big)\, f_c(r_{ij})\, Y_{\ell m}(\hat{\mathbf{r}}_{ij})$. Here T_n are Chebyshev polynomials ($n < 8$), $f_c(r) = \frac{1}{2}[\cos(\pi r/r_{\mathrm{cut}}) + 1]$ is SNAP's switching function, and $Y_{\ell m}$ are real spherical harmonics with $\ell \le 6$. The harmonics are evaluated as polynomials in (x, y, z), so they are smooth everywhere. The primitive expansion of atom i is $\tilde{A}^i_{n\ell m} = \sum_{j\in\mathcal{N}_i} g_{n\ell m}(\mathbf{r}_{ij})$. It contains no parameters and is the same object that underlies SOAP, ACE and NEP-type descriptors [2, 6–8].

Learned radial channels. For each ℓ, a learned matrix mixes the 8 radial functions into 12 channels, $A^i_{c\ell m} = \sum_n W^{(\ell)}_{cn}\,\tilde{A}^i_{n\ell m}/s_{n\ell}$. The fixed scale $s_{n\ell}$ is the root-mean-square norm of the (n, ℓ) block over the included atoms of the draw. When the included set is too large for one episode, a random subset of up to 256 included configurations is used.

Invariant contractions. From A we form three kinds of rotation- and inversion-invariant features. The two-body densities are A^i_{c00}. The three-body power spectrum $P^i_{cc'\ell} = \sum_m A^i_{c\ell m} A^i_{c'\ell m}$ couples every pair of channels $c \le c'$ at each $\ell \le 6$. The four-body, bispectrum-type invariants

$$Q^i_{c,\ell_1\ell_2\ell_3} = \sum_{m_1 m_2 m_3} C^{\ell_1\ell_2\ell_3}_{m_1 m_2 m_3}\, A^i_{c\ell_1 m_1} A^i_{c\ell_2 m_2} A^i_{c\ell_3 m_3} \tag{3}$$

are formed within each channel for the parity-even, triangle-admissible triples with $1 \le \ell_1 \le \ell_2 \le \ell_3 \le 4$. The coupling tensors C are obtained numerically as the one-dimensional rotation-invariant subspace of $V_{\ell_1} \otimes V_{\ell_2} \otimes V_{\ell_3}$ in our real basis. Restricting to even $\ell_1 + \ell_2 + \ell_3$ makes all features invariant under improper rotations as well. The descriptor has 666 invariants in total.

Nonlinear compression. The invariants are standardised with constants fixed from the included atoms. A multilayer perceptron (2 hidden layers of width 128, SiLU activations) then maps them to $K' = 16$ output channels, which form $\mathbf{B}^i_\theta$. Subtracting $\mathbf{B}_\theta(\varnothing)$, the output for an empty neighbourhood, gives $\Delta\mathbf{B}_\theta = \mathbf{0}$ for an isolated atom. Together with $\beta_0 = 0$ this reproduces `bzeroflag=1` exactly. The quadratic head then has $K' + K'(K'+1)/2 = 152$ coefficients, 10.5 times fewer than quadratic SNAP. The descriptor has 104,624 trainable parameters.

Invariances. Translation invariance holds because only relative vectors $\mathbf{r}_{ij}$ enter. Permutation invariance holds because neighbour contributions are summed and all atoms share one map. Rotation and inversion invariance hold because only contracted invariants reach the MLP. All three are exact algebraic properties of the architecture, as for the bispectrum. Section 5.4 tests them numerically, together with a deliberately non-invariant variant.

3.2 Bilevel least-squares training

The head is solved, not learned. For a set S of configurations, the design matrix $\mathbf{A}_S(\theta)$ of Eq. (2) has the same rows as FitSNAP's: one per-atom energy row and $3N$ force rows per configuration. The columns hold configuration sums of $\phi(\Delta\mathbf{B}^i_\theta)$ and their negative position derivatives. The force rows are assembled analytically. The per-atom Jacobian $\partial\Delta\mathbf{B}^i_\theta/\partial\tilde{A}^i$ is computed in closed form as the product of the MLP Jacobian, the derivatives of the power-spectrum and bispectrum contractions, and the linear radial mixing. It agrees with reverse-mode differentiation to 10^{-14} and is about twice as fast. It is then contracted with the fixed derivatives $\partial g/\partial\mathbf{r}_{ij}$ in one batched matrix product per centre atom, and the result is scattered to atoms i and j. The quadratic columns follow from $\partial(\Delta B_k\Delta B_l) = \Delta B_l\,\partial\Delta B_k + \Delta B_k\,\partial\Delta B_l$. Forces are therefore exact gradients of the energy, and agree with central finite differences to within 4×10^{-8} eV/Å (`code/blade/test_core.py`). $\mathbf{A}_S$ is differentiable with respect to θ.

Episodes: the cross-validated error of the head as the training objective. Training uses only the included configurations I of a draw. For Ta, an episode is the whole of I if it contains at most 4,096 atoms; otherwise it is a random subset of I up to that atom budget. For the transfer metals, whose datasets are an order of magnitude larger (Table 1), episodes are random 2048-atom subsets. This is a compute-driven choice, disclosed in Sec. 4; the architecture, objective, step count and schedule are identical across materials. At every step the episode's configurations are dealt into $k\sim\mathcal{U}\{2,\dots,4\}$ folds by standard stratified assignment: the configurations of each group are distributed round-robin, in random order and from a random starting fold, so every fold holds a share of every group large enough to be split. For each fold f the head is solved in closed form on all *other* folds,

$$\mathbf{c}_{-f}(\theta) = \big(\mathbf{G}-\mathbf{G}_f+\lambda\mathbf{I}\big)^{-1}\big(\mathbf{b}-\mathbf{b}_f\big), \qquad \mathbf{G}_f=\mathbf{A}_f^\top\mathbf{A}_f,\;\; \mathbf{b}_f=\mathbf{A}_f^\top\mathbf{y}_f, \tag{4}$$

where $\mathbf{A}_f(\theta)$ holds the column-scaled rows of fold f. The term $\lambda = 10^{-8}\times$ the mean diagonal is a numerical safeguard for the Cholesky factorisation, not a regulariser. Every configuration is then predicted *out of fold*, and the descriptor is updated on the group-balanced mean squared out-of-fold error

$$\mathcal{L}(\theta) = \frac{1}{|\mathcal{G}|}\sum_{g\in\mathcal{G}}\overline{\big(r^E_g\big)^2} + \frac{1}{|\mathcal{G}|}\sum_{g\in\mathcal{G}}\overline{\big(r^F_g\big)^2}, \qquad r=\mathbf{A}_f\mathbf{c}_{-f}-\mathbf{y}_f \text{ for the fold } f \text{ holding each row}, \tag{5}$$

where the bars average the energy rows (eV/atom) and the force rows (eV/Å) of training group g in the episode, and $\mathcal{G}$ is the set of groups present. Each group therefore contributes equally to what the descriptor learns, mirroring the equal-weight aggregate used for evaluation, while the least-squares head inside the loop and at deployment remains unweighted. Without this balancing, the three Liquid configurations of Ta (0.8 % of the energy rows) were systematically underfitted; Appendix A has the development record. In words, the descriptor minimises the k-fold cross-validated error of its own least-squares head. The expensive object, the design matrix of the episode, is built once per step. The k solves involve only $K'(K'+3)/2$-dimensional systems, so the cross-validation costs almost nothing. Gradients flow through the solves. Optimisation uses AdamW (learning rate 0.005, weight decay 10^{-5}, 50 warm-up steps, cosine decay to 1 %) for 1,500 steps with gradient clipping at norm 10; the last iterate is kept.

The descriptor is never trained to fit the configurations its head was solved on. It is trained so that a least-squares head solved on part of the data predicts the rest. This is the episodic, closed-form-head principle of few-shot learning [9, 10], here with the exact SNAP head, energies and forces, and a stratified cross-validation inner loop. It can also be read as variable projection [11] with a held-out rather than in-sample objective.

Final head and evaluation. After training, the design matrix of all configurations is formed with the trained descriptor and passed to the same FitSNAP SVD call used for quadratic SNAP, on the rows of I only. The two methods therefore differ only in the columns of $\mathbf{A}$.

Hyper-parameters and leakage control. All values above were fixed before the CV sweeps (`logs/preregistration.md`). They were chosen on Ta with two development signals only: the full-data training error, and internal 80/20 splits *inside* the included configurations of the $f=0.5$ and $f=0.1$ draw-0 sets. The development runs are listed in Appendix A, which also discusses what this

Table 1: Datasets. Configurations, atoms and force components after duplicate removal; n_g: number of training groups; subset: low-symmetry groups (Ta: GSF, Liquid, Surface) or defect groups (transfer metals: vacancy, interstitial). r_{cut} and reference potential are shared by quadratic SNAP and BLADE (Ta: paper values; Mo/Nb/Re: selected by the paper's criterion, `logs/rcut_scan`).

	configs	atoms	forces	n_g	subset		atoms/config	E/N range (eV)		r_{cut} (Å)	ref.
					groups	configs		min	max		
Ta	363	4,224	12,672	12	4	54	2–100	-11.85	28.13	5.594	ZBL
Mo	3,633	65,980	197,940	11	6	562	7–32	-10.75	18.94	6.80	none
Nb	3,093	53,858	161,574	15	6	574	1–32	-10.20	42.30	5.20	none
Re	5,017	101,000	303,000	10	6	1,597	7–32	-12.29	7.77	4.35	none

selection can and cannot leak. The same values were then used unchanged for every draw and every material, with the single exception of the episode size noted above. Each CV draw trains its own descriptor from scratch on its included configurations, with the draw's seed; normalisation constants also come from the included atoms only.

4 Data, baselines and protocol

Tantalum. We use the 363 DFT configurations of the original study without modification. They are divided into 12 groups: Displaced A15/BCC/FCC, Volume A15/BCC/FCC, Elastic BCC/FCC, GSF(110), GSF(112), Liquid and Surface (Table 1). The provided files contain 4,224 atoms, that is 12,672 force components. The paper's text quotes 15,297 force components, and the per-group counts differ for the Elastic, Liquid and Surface groups (§V of Ref. [4]). We return to this difference when we compare our reimplementation with the published numbers.

Transfer metals. Molybdenum, niobium and rhenium are taken from the curated single-element training sets of Andolina and Saidi [5, 12]. These are VASP ab-initio MD frames of several crystal polymorphs and of vacancy and interstitial supercells, sampled at several temperatures up to the melting point over several active-learning iterations. We convert the DeePMD arrays of the pre-staged archives to FitSNAP's JSON format, one configuration per DFT frame. We define training groups as parent structure $\times$ configuration family. The parent structures are identified with `spglib`: BCC, FCC, double-hexagonal and simple-hexagonal/cubic polymorphs for Mo; BCC, FCC, HCP, simple cubic and a low-symmetry phase for Nb; HCP, simple cubic and three further polymorphs for Re. The families are elastic, vacancy, interstitial and ground state. Frames that are exact duplicates of another frame (same positions, cell and energy) are removed. Otherwise the same DFT calculation would appear on both sides of a CV split. This removes 94, 153 and 403 frames. No other frame is filtered. These sets are far harder than the Ta set: they contain strongly compressed high-temperature frames, with nearest-neighbour distances down to 0.69 Å (Re), 0.96 Å (Mo) and 1.13 Å (Nb), and per-atom energies spanning 20–52 eV/atom. The DeePMD models that the data providers trained on them end with training energy RMSEs of 0.07–0.13 eV/atom and force RMSEs of 0.33–0.48 eV/Å (`lcurve.out` files shipped with the data).

Reference potential and cutoff. For Ta we keep the published choice: a ZBL reference potential switched off between 4.0 and 4.8 Å, $J_{\max} = 4$ and $r_{\mathrm{cut}} = 5.594$ Å (Table SII of Ref. [4]). For each transfer metal we apply the paper's own selection criterion (§V of Ref. [4]): the unweighted full-data training energy error of quadratic SNAP. We minimise it over $r_{\mathrm{cut}} \in [4.0, 7.0]$ Å and over the reference potential (ZBL or none), then refine r_{cut} on a 0.05 Å grid. The learned descriptor always uses the same cutoff and reference as the quadratic-SNAP baseline of that material.

Two quadratic-SNAP baselines. *Paper-as-reported*: $\delta E = 0.00898$ eV/atom for $J_{\max} = 4$, $r_{\mathrm{cut}} = 5.594$ Å (Table SII of Ref. [4]). It was obtained with unweighted least squares ($\gamma_n = 0$, $\mathbf{w} = \mathbf{1}$; §V of Ref. [4]). *Reimplemented*: FitSNAP [13] computes the bispectrum design matrix once per material with LAMMPS `compute snap` [14]. Every CV draw is then solved with FitSNAP's own SVD solver (`scipy.linalg.lstsq`, `cond`$= 10^{-13}$) on unweighted per-atom-energy and force rows, the same protocol as Table SII. As a check of the data path, the same code reproduces the

metrics shipped with FitSNAP's example of this potential exactly when that example's weights are used ($\delta E = 0.000238$ eV/atom and $\delta F = 0.01585$ eV/Å with energy weight 100).

Controls. Further least-squares models, all fitted with the same rows and solver, isolate where any gain comes from. Linear SNAP (55 coefficients) is the original paper's lower baseline. Our unweighted reimplementation does not reproduce its Table SI energy error, whereas it does reproduce the quadratic model; Appendix D traces this to the relative weight of energy and force rows, to which only the linear model is sensitive. On Ta we also fit quadratic SNAP with Tikhonov regularisation, whose strength is selected per draw by 5-fold cross-validation *inside* the included configurations. This is the classical remedy for overfitting. Finally, the quadratic head is fitted on an uncentred principal-component projection of the bispectrum onto K' channels, $\mathbf{z} = \mathbf{P}\Delta\mathbf{B}$, with $\mathbf{P}$ computed from the included atoms. This control has exactly as many head coefficients as the learned descriptor but no learning. Because $\sum_i z_a^i z_b^i$ is a fixed linear map of the quadratic-SNAP columns, its design matrix is computed exactly from FitSNAP's. Section 6.1 adds a fourth control, in which $\mathbf{P}$ itself is trained with BLADE's objective, and ablations of BLADE's training procedure and invariance.

Cross-validation protocol. We reproduce the protocol of Fig. 5 of Ref. [4]. The included fraction is $f = 0.05, 0.10, \ldots, 0.95$, with 10 random draws per fraction, plus the full-data point $f = 1$. Draws are stratified within every group. Group g with n_g configurations contributes $\min(\max(\lfloor f n_g + \frac{1}{2} \rfloor, 1), n_g - 1)$ included configurations, so every group has included and held-out data at every fraction. The energy and all forces of a configuration are always on the same side of a split. The 191 draws of each material are generated once from fixed seeds and shared by all methods, so every comparison is paired. All training-time decisions of the learned descriptor (normalisation constants, episodes, model selection) use only the included configurations of the draw.

Metrics. δE is the mean absolute error of the energy per atom and δF that of the Cartesian force components, on held-out configurations (on the training set at $f = 1$, the paper's full-data CV endpoint). For every draw we record pooled errors, per-group errors, the equal-weight aggregate over groups, and the aggregate over a pre-declared low-symmetry subset. For Ta that subset is GSF(110), GSF(112), Liquid and Surface. For the transfer metals it is the vacancy and interstitial groups. Curves show mean $\pm$ standard deviation over the 10 draws.

Data-efficiency statistic. The reference accuracy is the reimplemented quadratic SNAP's full-data error of the same aggregate. We define $F^\star$ as the smallest fraction at which the 10-draw mean held-out error of a method is at or below the reference *and stays there* for every larger fraction (a sustained crossing). This reference is a training error. We therefore also report a conservative variant, $F^\star_{95}$, whose reference is quadratic SNAP's own held-out error at $f = 0.95$. Both statistics are computed identically for all materials, for the equal-weight aggregate and for the subset, and were fixed before any sweep was run (`logs/preregistration.md`).

Invariance audit. For each material, the models trained on draw $f = 0.80$, draw 0 (the 80/20 split) are probed on 10 configurations drawn at random from that draw's held-out set. Each configuration is subjected to 100 uniformly random proper rotations of positions and cell (unit quaternions), to 10 rigid translations with components drawn from $\mathcal{U}[-1, 1]$ Å, and to 10 random permutations of the atom indices. For every transformed configuration we compare the predicted total energy and forces with the correspondingly transformed predictions for the original configuration ($\mathbf{RF}$, $\mathbf{F}$, $\mathbf{F}_\pi$). We report, as maxima over all samples, the relative energy drift $|\Delta E/E|$, the relative force drift $|\Delta F/F| = \max_a \|\Delta\mathbf{F}_a\| / \max(\max_a \|\mathbf{F}_a\|, 0.01\,\text{eV/Å})$, and the absolute force drift $\max_a \|\Delta\mathbf{F}_a\|$. The force metric is normalised by the largest force of the configuration, with a floor, because the forces of perfect crystals vanish and a per-atom ratio would divide by zero. We also report the per-atom energy drift $|\Delta E|/N$, which, unlike $|\Delta E/E|$, does not depend on the arbitrary zero of the energy scale.

BLADE is evaluated directly on the transformed coordinates, with its own neighbour list and no re-orientation, in double and in single precision. Quadratic SNAP is evaluated with LAMMPS `pair_style snap`. LAMMPS requires a restricted-triclinic cell, so rotated cells are mapped back to that frame by the same QR re-orientation that FitSNAP applies to all training data, and the forces are rotated back. For SNAP, the audit therefore measures the drift of the complete LAMMPS pipeline.

Inference cost. We time NVE molecular dynamics of BCC Ta supercells with 128, 1024 and 8192 atoms, with velocities drawn at 300 K on the perfect lattice, a 1 fs time step and a Verlet list with 1 Å skin. Quadratic SNAP runs in LAMMPS (22 Jul 2025, `pair_style hybrid/overlay zbl 4.0 4.8 snap`) on 1 and 16 MPI ranks. BLADE runs as a PyTorch velocity-Verlet integrator with forces obtained by one reverse pass through the descriptor, on one GPU (double and single precision) and on 1 or 16 CPU threads. We report wall-clock time per atom-step after warm-up, measured for both codes back to back on the otherwise idle node after all sweeps had finished. Energy conservation is checked separately in 10 ps NVE runs at about 300 and 2000 K.

Compute. All learned-descriptor training, audits and GPU timings ran on a single NVIDIA H200 GPU of a shared node. A full BLADE sweep is 191 independent trainings. To keep the three transfer-metal sweeps within this budget, their training episodes are capped at 2048 atoms instead of the 4096 used for Ta (Sec. 3.2). This is the only material-dependent training setting. Two or three trainings shared the GPU at any time. CPU work (LAMMPS descriptors, least-squares solves, LAMMPS timings) used at most 16 cores at a time, apart from transient overlap between CPU jobs and the GPU-driving processes.

Reproducibility of a single training run. BLADE's training is not bitwise reproducible. Some GPU operations we use, notably scatter-additions, are non-deterministic, and their round-off differences grow during training. When a resumed sweep inadvertently re-ran 48 Nb draws and 11 draws of a Ta ablation with identical seeds, the held-out equal-weight energy error differed between the two runs by a median of 12 % (Nb) and 26 % (Ta), and the held-out force error by 3–7 %. We report the first run of each draw; the re-runs are kept in `logs/cv_duplicates/`. All conclusions therefore rest on means over ten draws, which absorb this run-to-run variability, and the full-data fits are repeated with five seeds.

5 Results

5.1 Accuracy at full data and on the 80/20 split

Table 2 collects the headline numbers.

The reimplementation reproduces the published model. Our FitSNAP reimplementation of quadratic SNAP reaches a full-data training error of $\delta E = 0.0082$ eV/atom, against 0.00898 eV/atom reported in Table SII of Ref. [4]. The 8 % gap is discussed in Sec. 6.2.

Full data. With the identical least-squares head, BLADE fits the full Ta set to $\delta E = 0.0075$ eV/atom, slightly below both baselines, with 10.5× fewer head coefficients. Unlike the deterministic least-squares fit, BLADE's training is stochastic, so we repeated the full-data training with four further seeds (Table 2). The energy error then ranges from 0.0026 to 0.0075 eV/atom (mean 0.0042 ± 0.0017 over 5 seeds) and lies below both baselines in every run; the pre-registered run is the least accurate of the five. BLADE does not match quadratic SNAP on full-data *forces* in any run: $\delta F = 0.0267$–0.0364 against 0.0150 eV/Å. The full-data endpoint is a training error, and quadratic SNAP's 1595 coefficients fit the training forces more tightly.

The 80/20 split. On held-out configurations the ordering reverses for both quantities. At $f = 0.80$, BLADE's held-out errors are $\delta E = 0.0155 \pm 0.0051$ eV/atom and $\delta F = 0.0595 \pm 0.0050$ eV/Å. Quadratic SNAP's are 1.07 ± 1.21 and 0.0915 ± 0.0141. Accuracy parity therefore holds on the 80/20 protocol for both energies and forces. At the full-data endpoint it holds for energies only.

5.2 Data efficiency on tantalum

Figure 1 reproduces the cross-validation experiment of the original paper for every method. Quadratic SNAP shows the behaviour Wood and Thompson describe. Below $\sim$30% of the data its held-out errors are those of an underdetermined fit, with equal-weight averages of 9.0×10^5 eV/atom at $f = 0.05$. Even at $f = 0.95$ the equal-weight held-out energy error is 3.90 eV/atom, orders of magnitude above its own full-data training error. BLADE's held-out errors are smooth across the whole sweep. Its mean equal-weight held-out energy error is lower than quadratic SNAP's at 19/19 fractions, by a

Table 2: Primary results. δE (eV/atom) and δF (eV/Å) are pooled mean absolute errors. Full data: training error of the fit to all configurations (the paper's full-data endpoint, Table SII protocol). 80/20: held-out error at $f = 0.80$, mean $\pm$ std over 10 draws. $F^\star$: smallest included fraction from which BLADE's mean held-out δE stays at or below the reimplemented quadratic SNAP's full-data δE of the same aggregate (equal-weight over all groups / low-symmetry or defect subset; >0.95: no sustained crossing). MD cost: wall-clock time per atom and step for 128-atom BCC Ta relative to LAMMPS quadratic SNAP on the same 16 CPU cores, for BLADE on 16 CPU threads / on one GPU (Table 8). "training seeds": mean and, in parentheses, standard deviation in units of the last digit, of the full-data training errors over the pre-registered and four further training seeds of the descriptor. Lower-case rows: controls. Paper value: Wood and Thompson (2018), Table SII, row $J_{\max} = 4$.

	full data		80/20 split (held out)		$F^\star$	MD cost
Method	δE	δF	δE	δF	agg./subset	(rel.)
Ta: 363 configurations, 12 groups, $r_{\rm cut} = 5.594$ Å, ZBL reference						
Quadratic SNAP, paper (Table SII)	**0.00898**	–	–	–	–	–
Quadratic SNAP, reimpl.	0.0082	0.0150	1.07±1.21	0.0915±0.0141	ref.	1
BLADE-qSNAP	0.0075	0.0364	0.0155±0.0051	0.0595±0.0050	>0.95 / 0.55	6.9 / 5.1
5 training seeds	0.0042(17)	0.0328(33)	–	–	–	–
linear SNAP	0.152	0.0866	0.165±0.0348	0.104±0.0042	–	–
quadratic SNAP + ridge	0.0361	0.0329	3.99±4.79	0.205±0.196	–	–
PCA-16 bispectrum, quad. head	0.0783	0.1000	0.820±1.40	0.134±0.0088	–	–
Mo: 3,633 configurations, 11 groups, $r_{\rm cut} = 6.80$ Å, no reference						
Quadratic SNAP, reimpl.	0.0517	0.190	0.0521±0.0076	0.200±0.0089	ref.	–
BLADE-qSNAP	0.0641	0.213	0.0612±0.0080	0.216±0.0083	>0.95 / >0.95	–
Nb: 3,093 configurations, 15 groups, $r_{\rm cut} = 5.20$ Å, no reference						
Quadratic SNAP, reimpl.	0.0375	0.111	0.0408±0.0023	0.123±0.0157	ref.	–
BLADE-qSNAP	0.0383	0.157	0.0423±0.0076	0.170±0.0085	>0.95 / 0.05	–
Re (HCP): 5,017 configurations, 10 groups, $r_{\rm cut} = 4.35$ Å, no reference						
Quadratic SNAP, reimpl.	0.0233	0.221	0.0247±0.0025	0.235±0.0166	ref.	–
BLADE-qSNAP	0.0223	0.241	0.0225±0.0026	0.244±0.0056	>0.95 / >0.95	–
5 training seeds	0.0289(66)	0.241(4)	–	–	–	–

Table 3: Data efficiency. Reference: reimplemented quadratic SNAP's full-data δE of the same aggregate (training error) and, conservatively, its held-out δE at $f = 0.95$. Columns 5–6: 10-draw mean held-out δE at $f = 0.50$. $F^\star$, $F^\star_{95}$: smallest fraction from which BLADE's mean held-out δE stays at or below the respective reference. Bands follow the pre-registered definition (substantial: $F^\star \le 0.5$; modest: ≤ 0.8; matches: ≤ 0.95). All energies in eV/atom.

	aggregate	ref. (full)	ref. (f=0.95)	qSNAP f=0.5	BLADE f=0.5	$F^\star$	$F^\star_{95}$	band
Ta	equal-weight	0.0120	3.90	3.93	0.0284	>0.95	0.05	none
Ta	subset	0.0114	0.0249	0.0367	0.0137	0.55	0.25	modest
Mo	equal-weight	0.0464	0.0470	0.0499	0.0534	>0.95	>0.95	none
Mo	subset	0.0422	0.0461	0.0452	0.0487	>0.95	>0.95	none
Nb	equal-weight	0.0316	0.0313	0.0312	0.0332	>0.95	>0.95	none
Nb	subset	0.0336	0.0354	0.0306	0.0211	0.05	0.05	substantial
Re	equal-weight	0.0205	0.0215	0.0232	0.0196	>0.95	>0.95	none
Re	subset	0.0144	0.0160	0.0168	0.0163	>0.95	>0.95	none

factor 10^6 at $f = 0.05$, 138 at $f = 0.5$ and 175 at $f = 0.95$. Its held-out force error is lower at 19/19 fractions.

Data-efficiency statistics. Table 3 gives the pre-registered statistics. On the low-symmetry subset (GSF, Liquid, Surface), the discriminating test, BLADE's mean held-out energy error falls below quadratic SNAP's full-data value of 0.0114 eV/atom at $F^\star = 0.55$ and stays below it at every larger fraction. By the pre-registered bands this is a modest gain, just short of the substantial threshold of one half. Quadratic SNAP's own held-out subset error never reaches that value sustainably; it is 0.0249 eV/atom at $f = 0.95$. On the equal-weight aggregate, neither method reaches the strict reference (0.0120 eV/atom) on held-out data at any fraction. BLADE's lowest value is 0.0196, at



Figure 1: Tantalum cross-validation (Fig. 5 protocol of Ref. [4]). Held-out error versus included fraction; markers are means over 10 draws, bars $\pm$ one standard deviation (on the log axis the lower bar is truncated one decade below the mean). Open symbols at $f = 1$ are full-data training errors. The dashed line marks quadratic SNAP's full-data error, the reference for $F^\star$. Thin lines without bars are controls: linear SNAP, quadratic SNAP with inner-CV ridge, and the quadratic head on a 16-channel PCA of the bispectrum. Panels: (a, c) equal-weight average over all 12 groups; (b, d) the GSF(110), GSF(112), Liquid and Surface subset.

$f = 0.80$. The obstacle is the three Volume groups, analysed in Sec. 6.2: a held-out compressed or expanded crystal often lies outside the density range of the included configurations of its group. Against the conservative reference, quadratic SNAP's own held-out error at $f = 0.95$, BLADE reaches parity at $F^\star_{95} = 0.05$ for the aggregate and 0.25 for the subset. The aggregate value of this variant mainly reflects how badly quadratic SNAP extrapolates in density (its reference is 3.90 eV/atom); the subset value is the more informative one.

Per group: where BLADE does and does not win. Figures 2 and 3 resolve the twelve groups, and Table 10 lists the values. The aggregates hide a two-regime structure. While quadratic SNAP is underdetermined or barely determined ($f \leq 0.25$), BLADE has the lower held-out energy error in 52 of the 60 group–fraction cells; the exceptions are the energies of the two Elastic groups, which quadratic SNAP predicts better from $f = 0.10$ onwards. It has the lower force error in 35 of 35 cells. (The Volume and Elastic configurations are perfect, homogeneously strained crystals whose DFT forces vanish by symmetry, below 0.01 eV/Å, so their force errors are not counted.) Once quadratic SNAP is overdetermined ($f \geq 0.30$), it becomes the more accurate model on the near-crystalline groups. It has the lower energy error in 84 of the 98 Displaced, Elastic and GSF cells, by a median factor of 2.5 (at most 13), and the lower force error in 63 of the 70 Displaced and GSF cells, by a median factor of 1.6 (at most 2.3). These configurations are small perturbations of a few crystal structures, whose energies differ by a few meV/atom. Once a few dozen examples constrain the fit, 1595 coefficients on a fine fixed basis resolve these differences better than the learned descriptor, whose training signal is dominated by the groups with large errors (Sec. 6.2).

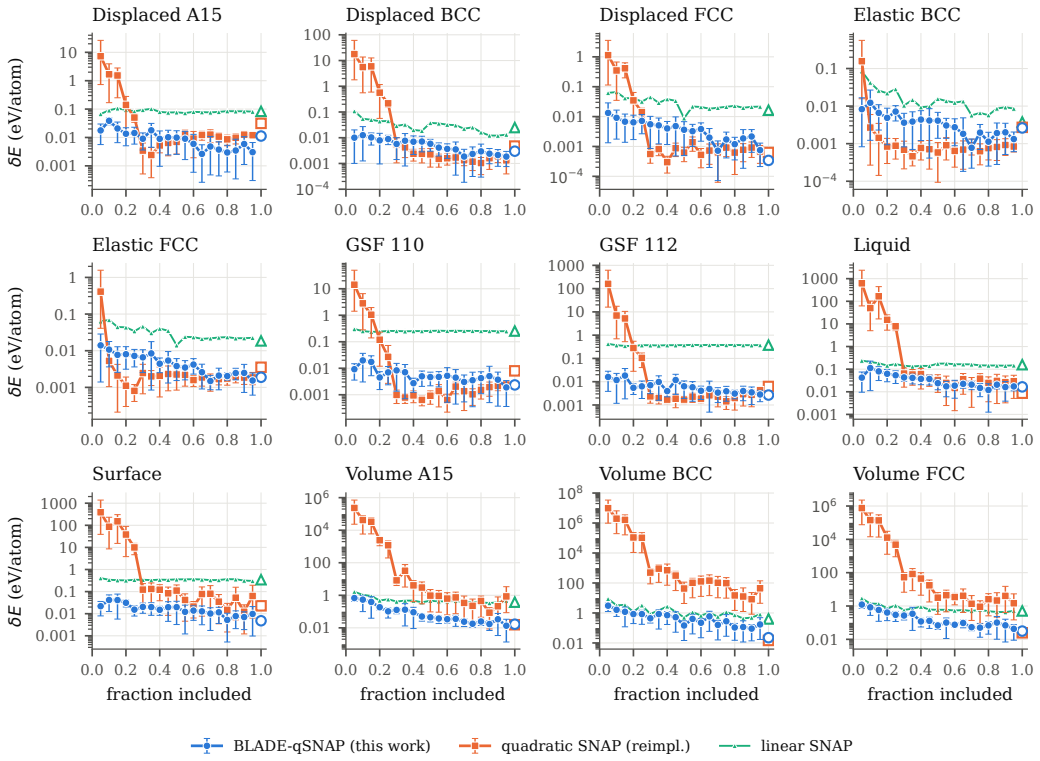


Figure 2: Per-group held-out energy errors on Ta (mean ± std over 10 draws; open symbols: full-data training error). Linear SNAP (thin line) is shown for orientation. Per-group force curves are in Appendix Fig. 6.

On the disordered and extrapolative groups the order is reversed. Over the same fractions, BLADE has the lower energy error in 68 of the 70 Liquid, Surface and Volume cells, by factors between 5.0 and 1,100 (median 62) on the Volume groups. It has the lower force error in 28 of the 28 Liquid and Surface cells, by a median factor of 2.6. These groups dominate the aggregates. There are only 3 Liquid configurations, but they hold 300 of the 4,224 atoms and the largest forces, and the Volume groups carry the largest energy errors. This is why BLADE wins the equal-weight and pooled aggregates at every fraction while losing most near-crystalline cells.

5.3 Transfer to molybdenum, niobium and rhenium

The transfer metals test the substitution in a different regime. Each dataset has thousands of configurations (Table 1), so even at $f = 0.05$ the included set provides at least 8,407 (Nb), 10,145 (Mo) or 15,389 (Re) least-squares rows, five to ten times the 1595 coefficients of quadratic SNAP, whereas the 22 Ta configurations at that fraction provide fewer rows than coefficients. The catastrophic small-data overfitting seen on Ta therefore cannot occur. What these sweeps measure is how each descriptor uses abundant data on strongly compressed, high-temperature configurations whose energies span more than ten eV/atom. Every BLADE model uses the Ta architecture, objective and schedule, with a descriptor trained from scratch on the included configurations of every draw (2048-atom episodes, Sec. 3.2). Figure 4 shows the curves, together with the training errors on the included configurations, and Table 2 gives the headline values. Per-group curves and win maps are in Appendix B.

Rhenium. Rhenium, the only HCP metal, is the transfer case closest to parity. On the 80/20 split, where every number averages ten trainings, BLADE reaches energy parity (0.0225 ± 0.0026 against 0.0247 ± 0.0025 eV/atom) but not force parity (0.244 against 0.235 eV/Å). At full data the pre-registered training run is slightly ahead in energy (0.0223 against 0.0233), but this is not robust. Over 5 training seeds the full-data energy error is 0.0289 ± 0.0066 eV/atom (range 0.0223–0.0414),

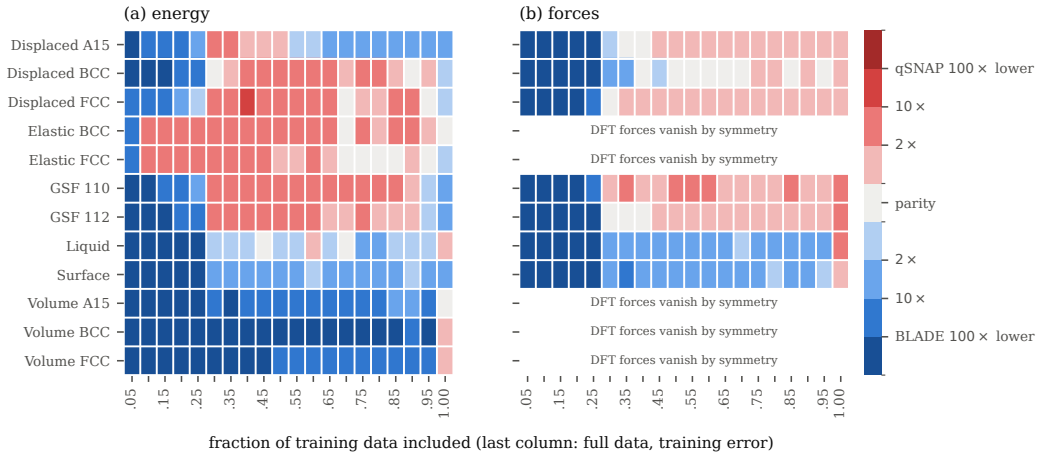


Figure 3: Where each descriptor wins on Ta. Each cell is $\log_{10}$ of the ratio of 10-draw mean held-out errors, quadratic SNAP over BLADE, for one group at one included fraction. Blue cells: BLADE has the lower error; red cells: quadratic SNAP does; grey: within a factor of 1.26 ($|\log_{10}| < 0.1$). The last column compares full-data training errors. Force cells are left blank for groups whose DFT forces vanish by symmetry.

and only 1 of the 5 runs is below quadratic SNAP's. The full-data force error, 0.235–0.245 eV/Å, is above quadratic SNAP's 0.221 in every run. Its held-out energy error is almost independent of the amount of data. The equal-weight mean stays between 0.0178 and 0.0235 eV/atom over the whole sweep, and its training error is as flat and nearly equal to it. At $f = 0.05$ BLADE's error is half of quadratic SNAP's (0.0189 against 0.0405). Quadratic SNAP's error falls steadily with more data and first matches BLADE's at $f = 0.85$. BLADE's mean lies below quadratic SNAP's full-data training error (0.0205 eV/atom) at 14/19 fractions, including the smallest. By the pre-registered sustained-crossing criterion, however, $F^\star > 0.95$: a flat curve that hovers at the reference crosses it repeatedly, and the three largest fractions, whose held-out sets are smallest and most variable, lie above it. On the vacancy and interstitial subset BLADE never reaches the reference (its lowest mean is 0.0147 against 0.0144), and quadratic SNAP is the more accurate model from $f = 0.55$ onwards. The per-group picture is mixed (Appendix Fig. 9). BLADE predicts energies better at every fraction on the HCP vacancy and both mp-1186901 groups, whereas quadratic SNAP is better on the simple-cubic vacancy, mp-8642 and mp-975065 groups once $f \geq 0.3$. Force errors are close everywhere. BLADE's are lower by more than 5 % in 91 of the 190 group–fraction cells and quadratic SNAP's in 40. BLADE has the lower equal-weight force error at 18/19 fractions, but the lower pooled force error at only 5/19. The pooled metric is dominated by the largest, high-force groups, on which quadratic SNAP is slightly more accurate.

Niobium. On niobium BLADE falls short of parity except on the defects. Its energy errors are close to quadratic SNAP's but slightly above them, both at full data (0.0383 against 0.0375 eV/atom) and on the 80/20 split (0.0423 ± 0.0076 against 0.0408 ± 0.0023). Its force errors are about 40 % higher (0.157 against 0.111 eV/Å at full data, 0.170 against 0.123 at 80/20). BLADE is ahead only at the smallest fractions: at $f = 0.05$ its equal-weight energy error is 0.0304 against 0.0449 eV/atom. Quadratic SNAP leads in the equal-weight energy from $f = 0.25$ and in pooled forces from $f = 0.10$. The defect subset behaves differently. There BLADE's held-out energy error is lower than quadratic SNAP's at all 19 fractions, by factors of 1.20 to 1.83 (0.0211 against 0.0306 eV/atom at $f = 0.5$). It lies below quadratic SNAP's full-data value already at $f = 0.05$, so the pre-registered statistic gives $F^\star = 0.05$, a substantial gain. The reference is soft, however. Quadratic SNAP's own held-out subset error at $f = 0.05$ (0.0224) is also below its full-data training value (0.0336). As configurations are added, the unweighted least-squares fit is pulled toward the much larger elastic groups, and quadratic SNAP's defect error grows from 0.0224 at $f = 0.05$ to 0.0354 at $f = 0.95$ (Appendix Fig. 11). BLADE's head is fitted by the same unweighted least squares, but its descriptor is trained with group balancing, and its defect error grows less, from 0.0171 to 0.0223. The robust statement is therefore the paired one: on Nb, BLADE predicts defect energies better than quadratic SNAP at every fraction

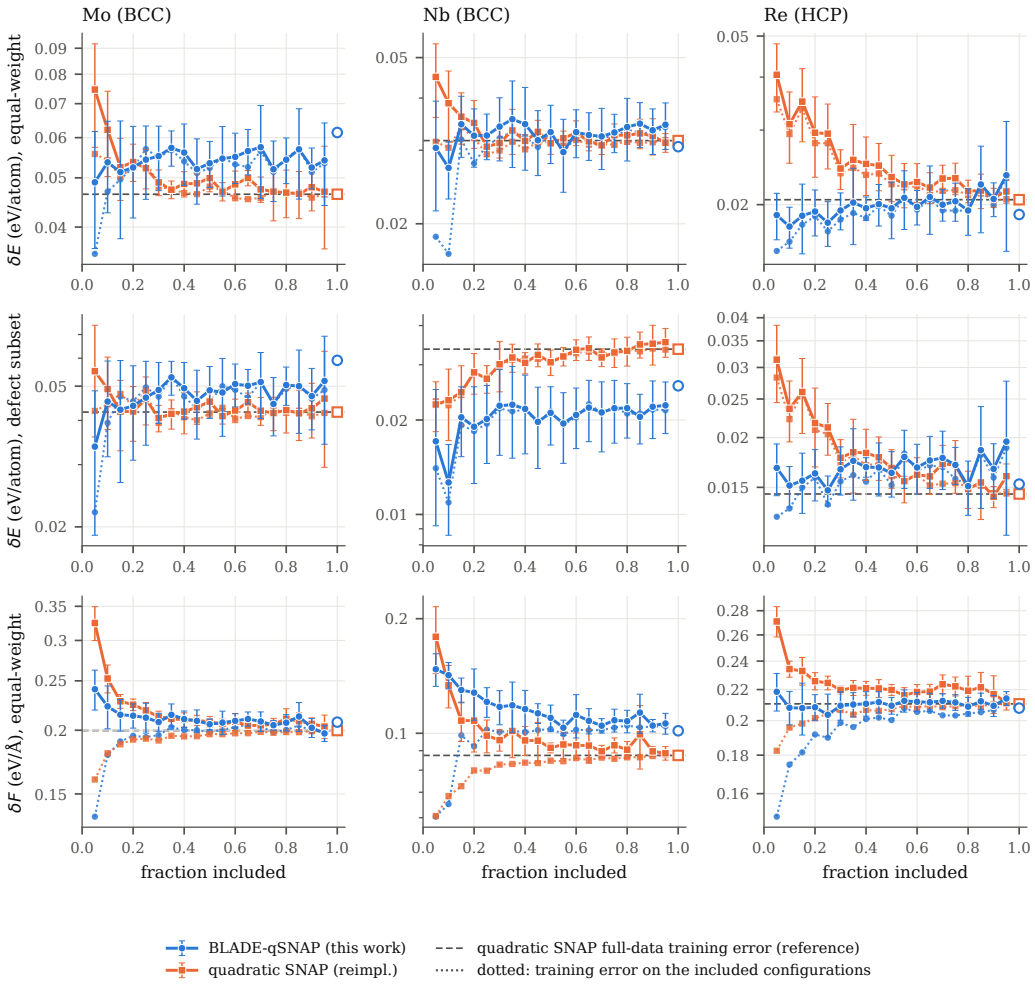


Figure 4: Transfer metals: held-out errors versus included fraction (mean $\pm$ std over 10 draws; solid lines), training errors on the included configurations (dotted lines, means), and full-data training errors (open symbols at $f = 1$). The dashed line is quadratic SNAP's full-data training error, the reference for $F^\star$. Rows: equal-weight energy over all groups; energy over the vacancy and interstitial groups; equal-weight force.

and, beyond the smallest fractions, everything else slightly worse. The win map (Appendix Fig. 10) shows the split seen on Ta. BLADE is better on the vacancy groups and worse on the perfect crystals ("ground") and the elastically deformed crystals, and quadratic SNAP's forces are better on most groups.

Molybdenum. Molybdenum, with the longest cutoff (6.80 Å, about 86 neighbours per atom), is the clearest regression. At full data BLADE's errors are 0.0641 eV/atom and 0.213 eV/Å, against 0.0517 and 0.190 for quadratic SNAP. On the 80/20 split they are 0.0612 ± 0.0080 against 0.0521 ± 0.0076 eV/atom and 0.216 against 0.200 eV/Å. BLADE is ahead at the smallest fractions, by a factor of 1.52 in the equal-weight energy at $f = 0.05$, but quadratic SNAP is more accurate in energy from $f = 0.25$ on, and neither aggregate reaches the reference ($F^\star > 0.95$). The vacancy groups are again the exception, this time in forces. BLADE's defect force error is lower at all 19 fractions, although its defect energy error is not. Per group (Appendix Fig. 7), BLADE predicts energies better at every fraction only on the simple-cubic vacancy group. Quadratic SNAP is better at almost every fraction on the FCC and DHCP vacancy groups and on the BCC elastic and interstitial groups.

Table 4: Post-hoc checks on the transfer metals, decided after the sweeps: a larger head ($K' = 32$) and, on Re, the 4096-atom episodes used for Ta (registered in `logs/preregistration.md` before they were run). "Sweep setting": $K' = 16$, 2048-atom episodes. Same splits and seeds as the sweeps; 10 draws per fraction (only $f = 0.5$ and full data for the 4096-atom episodes and for Mo with $K' = 32$). Errors in eV/atom and eV/Å; columns as in Table 6.

	full data (train)		$f = 0.10$ (held out)			$f = 0.50$ (held out)			$f = 0.80$ (held out)		
	δE	δF	E_{eqw}	E_{sub}	δF	E_{eqw}	E_{sub}	δF	E_{eqw}	E_{sub}	δF
Re											
BLADE-qSNAP (sweep setting)	0.0223	0.241	0.0178	0.0152	0.250	0.0196	0.0163	0.248	0.0194	0.0151	0.244
$K' = 32$	0.0280	0.229	0.0210	0.0175	0.243	0.0228	0.0179	0.237	0.0223	0.0175	0.228
4096-atom episodes	0.0205	0.238	–	–	–	0.0170	0.0144	0.243	–	–	–
quadratic SNAP	0.0233	0.221	0.0310	0.0236	0.264	0.0232	0.0168	0.236	0.0215	0.0149	0.235
Mo											
BLADE-qSNAP (sweep setting)	0.0641	0.213	0.0537	0.0452	0.251	0.0534	0.0487	0.219	0.0543	0.0504	0.216
$K' = 32$	0.0626	0.201	–	–	–	0.0619	0.0582	0.205	–	–	–
quadratic SNAP	0.0517	0.190	0.0622	0.0491	0.274	0.0499	0.0452	0.205	0.0468	0.0428	0.200

What limits BLADE on the transfer metals. Two observations explain the transfer results. First, BLADE's errors hardly depend on the amount of data, and they are not limited by overfitting. From $f = 0.25$ on, its equal-weight training error on the included configurations is within 7, 7 and 6 % of its held-out error on Mo, Nb and Re (dotted against solid lines in Fig. 4). At $f = 0.8$ its equal-weight training error is 0.0546 (Mo), 0.0336 (Nb) and 0.0194 eV/atom (Re), against held-out errors of 0.0543, 0.0341 and 0.0194. Quadratic SNAP is also close to its training error in this regime, but its error keeps falling as data are added. It crosses BLADE's equal-weight energy error at $f = 0.25$ on Mo and Nb and at $f = 0.85$ on Re, and on Mo it ends below BLADE's *training* error. With thousands of configurations, quadratic SNAP is limited by how much the bispectrum basis can express, BLADE by how well its 16 channels can be fitted within the training budget. Second, BLADE's remaining advantage is concentrated in the configurations farthest from equilibrium and in the defects (Table 11). More than 5 eV/atom above the ground state, its held-out energy error at $f = 0.5$ is lower than quadratic SNAP's on all three metals (0.0438 against 0.0716 eV/atom on Re, 0.157 against 0.186 on Mo). Within 1 eV/atom of the ground state it is comparable or higher. This is the pattern of Ta: the learned descriptor extrapolates more gracefully, and the fixed basis with many coefficients resolves near-equilibrium configurations better once enough of them are available.

Episode size and head size. The transfer sweeps use 2048-atom episodes. These hold only about 102 (Re) to 118 (Nb) configurations, that is, fewer energy rows than the head has coefficients, and the inner k-fold heads see only half to three quarters of them. The 4096-atom episodes of Ta hold all included configurations, up to 352. Two post-hoc checks (Table 4) support the reading that this limits the transfer results. With 4096-atom episodes on Re, the held-out equal-weight energy error at $f = 0.5$ falls from 0.0196 to 0.0170 eV/atom. The full-data error of this single training run is 0.0205, below quadratic SNAP's 0.0233 but within the seed spread of Re's full-data fits, while forces change little. Doubling the head instead ($K' = 32$) has the opposite effect on both Re and Mo. Held-out forces improve, on Mo to quadratic SNAP's level (0.205 against 0.205 eV/Å at $f = 0.5$), but held-out energies get worse (0.0619 against 0.0534 eV/atom). Energies are what the episodes constrain least, and a larger head leaves them even less determined. The compute-driven episode cap therefore has a real cost, and the transfer results understate what BLADE reaches with the episode size used for Ta.

5.4 Invariance audit

Table 5 reports the audit. BLADE's invariances are exact properties of the architecture, and in double precision the measured drifts are at the level of floating-point round-off for all three symmetries. Rotations change the total energy by at most 1.7×10^{-15} (relative) and the forces by at most 4.9×10^{-14} eV/Å; translations and permutations, which only shift or reorder sums of identical terms, give even smaller drifts. These values are below those of the quadratic-SNAP LAMMPS pipeline on the same configurations (5.3×10^{-14} and 1.4×10^{-13} eV/Å). For SNAP, the rotated cell has to be re-oriented into LAMMPS's restricted-triclinic frame. That involves an extra QR factorisation and a coordinate transformation in floating point; BLADE needs neither, because it accepts any cell orientation. In single precision the drift rises to 6.4×10^{-7} in relative energy and 7.8×10^{-6} eV/Å in

Table 5: Invariance audit on configurations held out from the $f = 0.80$, draw-0 split of each material, using the models trained on that split. "random": the protocol set of 10 randomly chosen held-out configurations; "force-carrying": a supplementary set of 10 held-out configurations with $\max|F_{\mathrm{DFT}}| > 0.5\,\mathrm{eV/Å}$, spread over the Displaced, GSF, Liquid and Surface groups. Entries are maxima over all samples of the relative total-energy drift $|\Delta E/E|$, the normalised force drift $|\Delta F/F| = \max_a \|\Delta\mathbf{F}_a\| / \max(\max_a \|\mathbf{F}_a\|, 0.01\,\mathrm{eV/Å})$ and, for rotations, the absolute force drift $|\Delta F| = \max_a \|\Delta\mathbf{F}_a\|$ in eV/Å. Transfer metals: models of their own 80/20 split, draw 0; per-sample values are in `logs/audit/`. BLADE-NI: the ablation that feeds the raw, rotation-equivariant density coefficients to the MLP; "+rot.aug.": the same, trained with a random rotation of every configuration at every step.

	rotation (1000 samples)			translation (100)		permutation (100)	
model	$\lvert\Delta E/E\rvert$	$\lvert\Delta F/F\rvert$	$\lvert\Delta F\rvert$	$\lvert\Delta E/E\rvert$	$\lvert\Delta F/F\rvert$	$\lvert\Delta E/E\rvert$	$\lvert\Delta F/F\rvert$
Ta, 10 random held-out configurations							
qSNAP (LAMMPS)	5.3×10^{-14}	1.4×10^{-11}	1.4×10^{-13}	1.8×10^{-14}	6.7×10^{-12}	5.5×10^{-15}	1.3×10^{-12}
BLADE fp64	1.7×10^{-15}	4.9×10^{-12}	4.9×10^{-14}	6.8×10^{-16}	1.0×10^{-12}	4.6×10^{-16}	5.3×10^{-13}
BLADE fp32	6.4×10^{-7}	5.2×10^{-4}	7.8×10^{-6}	7.3×10^{-7}	2.2×10^{-4}	5.5×10^{-7}	2.6×10^{-4}
BLADE-NI fp64	0.28	3.4	4.0	6.7×10^{-16}	8.6×10^{-13}	6.0×10^{-16}	8.2×10^{-13}
NI+aug. fp64	0.024	0.67	0.97	7.6×10^{-16}	1.0×10^{-12}	7.6×10^{-16}	1.2×10^{-12}
Ta, 10 force-carrying held-out configurations							
qSNAP (LAMMPS)	4.9×10^{-15}	4.7×10^{-13}	4.0×10^{-13}	3.5×10^{-15}	1.8×10^{-13}	2.0×10^{-15}	1.2×10^{-13}
BLADE fp64	5.5×10^{-16}	6.9×10^{-14}	7.0×10^{-14}	4.5×10^{-16}	3.5×10^{-14}	4.1×10^{-16}	1.0×10^{-14}
BLADE fp32	2.9×10^{-7}	1.7×10^{-5}	1.6×10^{-5}	2.9×10^{-7}	6.1×10^{-6}	2.9×10^{-7}	6.4×10^{-6}
BLADE-NI fp64	0.049	9.4	4.3	4.1×10^{-16}	4.5×10^{-14}	4.1×10^{-16}	1.0×10^{-14}
NI+aug. fp64	7.6×10^{-3}	2.0	2.0	4.1×10^{-16}	4.9×10^{-14}	4.0×10^{-16}	1.7×10^{-14}
Mo, 10 random held-out configurations							
qSNAP (LAMMPS)	4.4×10^{-15}	3.8×10^{-14}	1.6×10^{-13}	2.1×10^{-15}	2.0×10^{-14}	1.9×10^{-15}	4.8×10^{-15}
BLADE fp64	3.7×10^{-15}	2.7×10^{-14}	7.3×10^{-14}	9.2×10^{-16}	1.2×10^{-14}	1.1×10^{-15}	4.5×10^{-15}
BLADE fp32	4.2×10^{-7}	5.6×10^{-6}	2.4×10^{-5}	2.8×10^{-7}	2.4×10^{-6}	1.8×10^{-7}	2.3×10^{-6}
Nb, 10 random held-out configurations							
qSNAP (LAMMPS)	5.2×10^{-15}	4.4×10^{-14}	1.7×10^{-13}	3.1×10^{-15}	1.6×10^{-14}	1.9×10^{-15}	4.3×10^{-15}
BLADE fp64	1.6×10^{-15}	2.0×10^{-14}	7.2×10^{-14}	4.3×10^{-16}	6.3×10^{-15}	3.8×10^{-16}	7.0×10^{-15}
BLADE fp32	7.5×10^{-7}	6.2×10^{-6}	2.3×10^{-5}	3.7×10^{-7}	1.5×10^{-6}	3.7×10^{-7}	1.7×10^{-6}
Re, 10 random held-out configurations							
qSNAP (LAMMPS)	1.5×10^{-13}	4.3×10^{-14}	2.4×10^{-12}	1.0×10^{-13}	3.6×10^{-14}	1.6×10^{-14}	6.3×10^{-15}
BLADE fp64	7.5×10^{-14}	1.3×10^{-14}	9.1×10^{-13}	1.6×10^{-14}	1.0×10^{-14}	2.8×10^{-15}	1.8×10^{-15}
BLADE fp32	1.0×10^{-5}	6.5×10^{-6}	5.6×10^{-4}	4.7×10^{-6}	4.1×10^{-6}	3.3×10^{-6}	8.2×10^{-7}

forces (1.6×10^{-5} on the force-carrying set), as expected from the round-off of summing about 43 neighbour contributions and propagating them through a network of width 128. It is far below any physically relevant force, but a reason to run long, energy-conserving MD in double precision.

Transfer metals. The same audit on the 80/20 models of the transfer metals gives the same picture (Table 5). On Mo and Nb, BLADE's double-precision rotation drift is 3.7×10^{-15} and 1.6×10^{-15} in relative energy and 7.3×10^{-14} and 7.2×10^{-14} eV/Å in force, against 4.4×10^{-15} and 5.2×10^{-15} (1.6×10^{-13} and 1.7×10^{-13} eV/Å) for the LAMMPS pipeline. Single precision gives 4.2×10^{-7} and 7.5×10^{-7}, as on Ta. On Re, the HCP test case, BLADE's double-precision rotation drift is 7.5×10^{-14} in relative energy and 9.1×10^{-13} eV/Å in force, again no larger than that of the LAMMPS pipeline (1.5×10^{-13} and 2.4×10^{-12} eV/Å). In single precision the relative energy drift reaches 1.0×10^{-5}, more than on Ta, but this is an artefact of the metric rather than of the HCP environments. The maximum comes from one compressed HCP elastic configuration (16 atoms, nearest-neighbour distance 1.34 Å, forces up to 94 eV/Å) whose total DFT energy, 9.67 eV, happens to lie close to the arbitrary zero of the VASP energy scale. Its absolute drift, 6.6×10^{-6} eV/atom, is essentially the largest single-precision drift found on Ta (6.7×10^{-6} eV/atom); on the other nine Re configurations the relative drift is at most 3.4×10^{-7}. The large forces of this configuration likewise explain the larger absolute force drift. Per-atom energy drift is therefore the meaningful measure when total energies are referenced to an arbitrary zero.

What breaks when invariance is not built in. BLADE-NI removes only the invariant contraction: the 12×49 channel coefficients $A_{c\ell m}$ feed the MLP directly. Translation and permutation invariance

Table 6: Ablations and controls on Ta (means over the 10 draws at each of $f = 0.10, 0.50, 0.80$; every variant uses the splits and seeds of the headline sweep). Full data: training errors. E_{eqw}/E_{sub}: held-out energy error (eV/atom), equal-weight over all groups / GSF–Liquid–Surface subset; δF: pooled held-out force error (eV/Å). Variants of BLADE: (a) descriptor and a free head trained jointly by AdamW on the in-sample error, head refitted by least squares; (b) descriptor at its random initialisation, least-squares head only; (c) loss without group balancing; (d) raw, rotation-equivariant channel coefficients fed to the MLP; (e) as (d), with a random rotation of every configuration at every step; (f, g) other output dimensions; (h) the bispectrum compressed by a learned 16×55 projection trained with BLADE's objective. PCA-16 and quadratic SNAP: from their 191-draw sweeps.

	full data (train)		$f = 0.10$ (held out)			$f = 0.50$ (held out)			$f = 0.80$ (held out)		
variant	δE	δF	E_{eqw}	E_{sub}	δF	E_{eqw}	E_{sub}	δF	E_{eqw}	E_{sub}	δF
BLADE-qSNAP	0.0075	0.0364	0.272	0.0464	0.0602	0.0284	0.0137	0.0484	0.0196	0.0061	0.0595
(a) joint training	0.0082	0.0258	0.496	0.0185	0.0751	0.0587	0.0204	0.0572	0.0363	0.0254	0.0738
(b) untrained descriptor	13.8	0.681	10.5	9.86	0.849	12.2	12.1	0.735	11.4	11.5	1.17
(c) no group balance	0.0039	0.0271	0.325	0.0421	0.0557	0.0356	0.0336	0.0464	0.0269	0.0250	0.0602
(d) non-invariant (NI)	0.0076	0.0402	0.454	0.0413	0.216	0.0315	0.0285	0.101	0.0319	0.0127	0.153
(e) NI + rot. aug.	0.126	0.0970	0.210	0.0739	0.165	0.110	0.0335	0.0947	0.118	0.0408	0.114
(f) $K' = 8$	0.0018	0.0204	0.295	0.0303	0.0613	0.0242	0.0102	0.0451	0.0215	0.0071	0.0599
(g) $K' = 32$	0.0049	0.0253	0.448	0.0358	0.110	0.0327	0.0156	0.0527	0.0224	0.0081	0.0646
(h) learned projection	0.0569	0.0512	2.76	0.0775	0.0631	0.232	0.140	0.0512	0.256	0.121	0.0694
PCA-16 bispectrum	0.0783	0.1000	20.7	0.0945	0.102	1.01	0.0776	0.0959	1.15	0.0758	0.134
quadratic SNAP	0.0082	0.0150	1.8×10^{5}	36.4	233	3.93	0.0367	0.0785	1.49	0.0108	0.0915

survive exactly, because the neighbour sum and the relative coordinates are untouched, and both drifts stay at round-off. Rotation invariance is lost entirely. Rotating a held-out configuration changes the predicted energy by a median of 0.24 eV/atom (maximum 2.5 eV/atom) and individual forces by up to 4.0 eV/Å. In its training frame the same model has held-out errors of only 0.0286 eV/atom and 0.119 eV/Å. The orientation dependence is therefore one to two orders of magnitude larger than the error the model makes in its training frame: a model trained in one orientation is a different model in another. Rotation augmentation shows every training configuration in a fresh random orientation at every step. It reduces the energy drift sixfold in the median (to 0.041 eV/atom) and elevenfold in the maximum (to 0.22 eV/atom), but leaves force drifts of 2.0 eV/Å on the force-carrying set. It also pays for the approximate invariance with accuracy: its held-out energy error on the same split is 0.128 eV/atom, and even its training error is 0.116 eV/atom, because a network of fixed size must spend capacity learning a symmetry the invariant architecture never has to learn. The audit is therefore not a formality here. It separates a descriptor family whose invariance holds to machine precision from one whose errors are dominated by its residual orientation dependence. Section 6.1 quantifies what the missing invariance costs in held-out accuracy.

6 Mechanisms, failure modes and cost

6.1 Where the data efficiency comes from

Coefficient count is necessary but not sufficient. Quadratic SNAP's overfitting at small fractions is first a counting problem. At $f = 0.05$ the 22 included Ta configurations provide 1,222–1,294 least-squares rows for 1595 coefficients, and at $f = 0.10$ the 37 configurations provide 1,531–1,603, so the fit is underdetermined or barely determined. Beyond that, the absolute energy scale of the 1540 quadratic coefficients rests on few energy rows: 72 at $f = 0.2$ and 185 at $f = 0.5$. BLADE's head has 152 coefficients and is overdetermined at every fraction. The PCA-16 control has the same head size: it projects the bispectrum onto its 16 leading uncentred principal components and fits the same quadratic form. It is far worse than BLADE at every fraction tested (Table 6), and at full data it is about ten times worse than quadratic SNAP (0.0783 eV/atom). A small head alone buys stability, not accuracy.

What the training objective contributes, and what the learned features contribute. BLADE's channels are, to a good approximation, linear functions of the bispectrum. Over all 4,224 Ta atoms, ridge regressions predict each of BLADE's 16 standardised channels from the 55 bispectrum components with a cross-validated R^2 of 0.99 (median) and no channel below 0.97. Conversely, BLADE's 152 head features reproduce every bispectrum component with $R^2 \geq 0.90$. One might

conclude that BLADE merely selects a better 16-dimensional subspace of the bispectrum. To test this we trained such a subspace directly. A learned projection $\mathbf{z} = \mathbf{P}\Delta\mathbf{B}$ of the bispectrum is optimised with exactly BLADE's objective, starting from the PCA-16 projection. Its gradients flow through the same closed-form head, since the projected design matrix is a fixed bilinear function of $\mathbf{P}$ and of quadratic SNAP's columns.

The result separates two effects (Table 6). For forces, the projection's held-out errors ($0.0512\,$eV/Å at $f = 0.5$) are close to BLADE's (0.0484) and far below PCA-16's (0.0959). For energies, the projection lowers PCA-16's equal-weight error severalfold (0.232 against $1.01\,$eV/atom at $f = 0.5$) but remains far above BLADE's 0.0284, and its full-data training error is $0.0569\,$eV/atom. The cross-validated objective therefore accounts for most of BLADE's *force* generalisation even when the features are a linear compression of the bispectrum. The energy accuracy, by contrast, lives in the few per cent of channel variance that is *not* linear in the bispectrum. Energies of the groups that matter differ by meV/atom against feature variations of order one, so small nonlinear corrections to the descriptor are precisely what an energy fit needs.

Training is what makes sixteen channels sufficient. The same architecture at its random initialisation, with only the least-squares head fitted (variant b in Table 6), fails outright: its full-data training error is $13.8\,$eV/atom. The failure is not numerical. The design matrix of these random features is well conditioned (condition number 74, against 1.0×10^6 for the trained descriptor), and a fit to the 363 energy rows alone reaches $0.0003\,$eV/atom. What random channels cannot do is fit energies and forces with one set of coefficients: the combinations that reproduce the energies have position derivatives unrelated to the DFT forces. Training aligns the few channels with the directions in which energy and force data agree, which is also why the trained features are far more correlated with each other than random ones.

The out-of-fold objective, not joint training, generalises. Variant (a) trains the same descriptor jointly with a free head on the in-sample error of the included configurations, as an end-to-end network would be trained, and then refits the head by least squares exactly as BLADE does. It fits the full training set about as well as BLADE in energy (0.0082 against $0.0075\,$eV/atom) and better in force (0.0258 against $0.0364\,$eV/Å, mainly through the Liquid forces), but it generalises worse. Its held-out equal-weight energy error is about twice BLADE's at every fraction tested (0.496, 0.0587 and 0.0363 against 0.272, 0.0284 and 0.0196 at $f = 0.1$, 0.5 and 0.8), and its held-out force error is higher at every fraction (0.0572 against $0.0484\,$eV/Å at $f = 0.5$). At $f = 0.1$ the jointly trained descriptor reaches a lower *training* error than BLADE but a higher held-out error, the signature of in-sample overfitting that the out-of-fold objective is designed to prevent. The only aggregate in which joint training is better, at any of the four fractions, is the subset energy at $f = 0.1$ (0.0185 against $0.0464\,$eV/atom). There the included set holds a single Liquid and a single Surface configuration. In every episode these are predicted by a head that has seen no other configuration of their group, whereas the deployed head is fitted with them. For singleton groups, the out-of-fold objective therefore trains for a harder task than the one met at deployment.

Group balancing protects the hard groups. Without the group-balanced loss (variant c), the descriptor fits the full training set better ($\delta E = 0.0039$ against $0.0075\,$eV/atom, $\delta F = 0.0271$ against $0.0364\,$eV/Å) because it spends its capacity on the many near-crystalline rows. It underfits the three Liquid configurations, whose training energy error rises to $0.0760\,$eV/atom against 0.0165. On held-out data the cost falls where the subset criterion is measured. From $f = 0.3$ on, the subset energy error is 2.5 to 4 times BLADE's (0.0336 against $0.0137\,$eV/atom at $f = 0.5$, 0.0250 against 0.0061 at $f = 0.8$), and the equal-weight error is 20–40 % higher at every fraction. Pooled held-out forces change by less than 8 %. Balancing the descriptor's loss over groups, while keeping the head unweighted, is therefore what makes the subset result of Table 3 possible.

Invariance by construction is also a data-efficiency device. The non-invariant variant (d) feeds the 12×49 raw channel coefficients to the network. In the cross-validation it never meets a rotated configuration: FitSNAP places every cell in the same lattice-aligned frame, so training and held-out configurations of a group share their orientation. Even so, its held-out forces are 2 to 3.6 times worse than BLADE's at every fraction (0.101 against $0.0484\,$eV/Å at $f = 0.5$), and its equal-weight energy error is higher at every fraction, although its full-data training errors are close to BLADE's ($0.0076\,$eV/atom, $0.0402\,$eV/Å). The 588 orientation-dependent inputs give the network many more

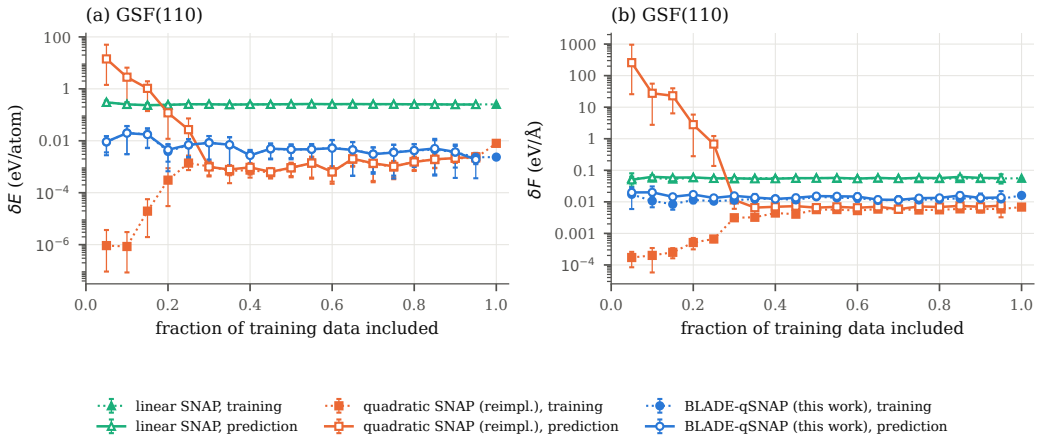


Figure 5: The analogue of Fig. 5 of Ref. [4] for GSF(110): training errors (closed symbols, dotted) and prediction errors on the held-out configurations (open symbols, solid) versus included fraction, mean ± std over 10 draws. Quadratic SNAP's training error collapses where its prediction error explodes. BLADE's two curves stay within a factor of two of each other at every fraction. Once quadratic SNAP is overdetermined it is the more accurate model on this group.

ways to fit the included forces. It fits the Liquid training forces five times better than BLADE (0.0307 against 0.157 eV/Å), but what it learns does not carry over to new configurations. Building in the symmetry removes these spurious directions before the data are seen, which is the same mechanism by which the bispectrum makes linear and quadratic SNAP data-efficient. Rotation augmentation (variant e) is no substitute. Showing every training configuration in a fresh random orientation at every step reduces the orientation dependence (Sec. 5.4), but within the same training budget it costs accuracy almost everywhere. The full-data training error rises to 0.126 eV/atom, about seventeen times BLADE's. From $f = 0.3$ on, the held-out energy errors are 2 to 6 times BLADE's (0.110 against 0.0284 eV/atom at $f = 0.5$), and the force errors are 1.9 to 2.7 times BLADE's at every fraction. Compared with the non-augmented variant, augmentation lowers the held-out force error by 6–26 % but raises the energy error more than threefold from $f = 0.5$ on. The network has to learn from data a symmetry that the invariant contraction provides exactly, and within 1,500 steps it does not.

Head size. The output dimension trades the variance of the head against the resolution of the descriptor (variants f and g). With $K' = 32$ the head has 560 coefficients, and the held-out errors are higher than with $K' = 16$ in eleven of the twelve aggregates at the four ablation fractions. The damage is largest at $f = 0.1$, where 560 coefficients face 1,531–1,603 rows: held-out forces are 0.110 against 0.0602 eV/Å. With $K' = 8$ (44 coefficients) BLADE is better than with $K' = 16$ in six of the twelve and never more than 16 % worse. Halving the head therefore costs nothing on held-out data, and doubling it overfits. The frozen value $K' = 16$ was chosen on development splits before these ablations and is not the optimum on Ta; we report it as frozen. On the data-rich transfer metals a larger head behaves differently, improving forces but not energies (Sec. 5.3).

Fitting versus predicting. Quadratic SNAP fits training forces more tightly than BLADE (0.0150 vs 0.0364 eV/Å at full data) yet predicts held-out forces worse at every fraction. A model with 10.5× more linear coefficients on a fixed, generic basis memorises configuration-specific force detail. That detail does not transfer to unseen configurations of the same group. BLADE spends its capacity in the descriptor, under an objective that only ever measures held-out error. Its training error therefore never runs far ahead of its test error. The gap between full-data training error and held-out error at $f = 0.95$, a direct measure of overfitting, is $0.170/0.0150$ for quadratic SNAP against $0.0966/0.0364$ for BLADE (eV/Å). Figure 5 shows the same contrast on the group that the original paper used as its example. Quadratic SNAP's training and prediction errors are separated by more than 7 orders of magnitude in energy and 6 in force at the smallest fractions, and come within a factor of two of each

Table 7: Ta, held-out bulk configurations at $f \geq 0.30$ (all draws), split by whether the configuration's volume per atom lies inside (interpolation) or outside (extrapolation) the range of the included configurations of its group. Energy errors in eV/atom; last column: fraction of cases in which BLADE's error is lower.

group family	held-out config.	n	median $\lvert\Delta E\rvert/N$		mean $\lvert\Delta E\rvert/N$		BLADE better
			BLADE	qSNAP	BLADE	qSNAP	(fraction)
Volume	interpolation	3697	0.033	0.081	0.102	14.3	0.74
Volume	extrapolation	563	0.190	5.35	0.787	668.2	0.93
Elastic	interpolation	10050	0.0025	0.0008	0.0038	0.0013	0.23
Elastic	extrapolation	450	0.0041	0.0015	0.0055	0.0035	0.33

other only from $f = 0.30$ (energy) and $f = 0.40$ (forces). BLADE's two curves never differ by more than a factor of 1.9, the signature of a model selected by its own cross-validated error.

6.2 Failure analysis

Reimplementation gap. Our quadratic SNAP reaches 0.0082 eV/atom against the published 0.00898 eV/atom at identical $J_{\max}$, cutoff and protocol, an 8 % difference in our favour. We trace it to the data rather than the fit, for two reasons. First, the paper's text counts 15,297 force components, while the provided configurations contain 12,672; the Liquid, Surface and Elastic counts differ. Second, the same pipeline reproduces the metrics shipped with FitSNAP's example of this potential exactly. Smaller contributions may come from the LAMMPS/FitSNAP versions (2024 vs. 2018 SNAP kernels). The reimplementation is slightly *better* than the published value, so the paired comparison does not flatter BLADE. Against the published number the full-data energy comparison is unchanged in sign. The linear-SNAP value of Table SI, in contrast, is not reproduced by any single convention for weighting energy and force rows that also reproduces Table SII (Appendix D). Linear SNAP is therefore only a qualitative control here.

Density extrapolation (Volume groups). Volume configurations are perfect crystals compressed or expanded isotropically, with per-atom energies up to 40 eV above equilibrium. Because ZBL contributes up to 87 eV/atom for these configurations, the descriptor part must cancel most of the reference repulsion (`logs/data/ta_target_ranges.json`). We label each held-out Volume configuration of every draw with $f \geq 0.30$ as an *extrapolation* when its volume per atom lies outside the range of the included configurations of its group, and as an *interpolation* otherwise (Table 7). When the held-out configuration is an interpolation, BLADE's median error is 0.033 eV/atom and quadratic SNAP's 0.081. When it is an extrapolation, BLADE's median error rises to 0.190, and quadratic SNAP's to 5.35. BLADE degrades far more gracefully: a smooth network on a Chebyshev radial basis, read out by 152 coefficients, extrapolates more tamely than 1540 quadratic coefficients on a fixed basis. It still fails, though, at the level that matters for the strict aggregate statistic. Held-out extrapolations are frequent at every fraction, because the random draws are stratified but not sorted in density. This is why no method reaches quadratic SNAP's full-data equal-weight error on held-out data. The remedy lies in the data or the physics, not the descriptor: the training set should span the density range of interest, or the reference repulsion should be retuned.

Near-crystalline groups with abundant data. On the Displaced, Elastic and GSF groups, quadratic SNAP predicts held-out energies more accurately than BLADE once it is overdetermined (Fig. 3). These are errors at the meV/atom scale, and head size does not explain them. With $K' = 8$ the descriptor fits every one of these groups at full data better than quadratic SNAP does, yet it remains less accurate than quadratic SNAP in 19 of their 21 held-out cells at $f = 0.3$, 0.5 and 0.8 (with $K' = 16$: 20; `logs/analysis/ta_group_spread.json`). What distinguishes these groups is how little their energies vary. On the Elastic groups and on Displaced A15 and FCC, the energy per atom of a configuration differs from its group mean by only 0.3–3.1 meV/atom on average. At $f = 0.5$ the held-out errors of both methods exceed this spread on most of these groups, so both err mainly in the absolute energy of each near-perfect crystal structure rather than in its variation with strain or displacement. BLADE's offsets (3.6–9.9 meV/atom) are larger than quadratic SNAP's (0.6–7.0 meV/atom). We attribute this to the training signal. Even with group balancing, the squared errors of these groups are one to three orders of magnitude smaller than those of the Volume, Liquid

Table 8: Molecular-dynamics cost, wall-clock seconds per atom and time step, for BCC Ta with velocities drawn at 300 K (NVE, 1 fs, Verlet list with 1 Å skin). Quadratic SNAP: LAMMPS `pair_style snap` with ZBL, our full-data reimplementation coefficients, MPI ranks on one node. BLADE-qSNAP: the full-data Ta model in PyTorch (energy and analytic forces in one reverse pass), on one GPU or on CPU threads. (`logs/timing/md_timing.jsonl`).

code	hardware	$N = 128$	$N = 1024$	$N = 8192$
quadratic SNAP (LAMMPS)	CPU 1 core	3.0×10^{-4}	2.9×10^{-4}	2.9×10^{-4}
quadratic SNAP (LAMMPS)	CPU 16 cores	2.7×10^{-5}	2.0×10^{-5}	1.9×10^{-5}
BLADE-qSNAP (PyTorch)	GPU fp64	1.4×10^{-4}	1.8×10^{-5}	3.1×10^{-6}
BLADE-qSNAP (PyTorch)	GPU fp32	1.4×10^{-4}	1.8×10^{-5}	2.8×10^{-6}
BLADE-qSNAP (PyTorch)	CPU fp64, 1 thread	2.4×10^{-4}	3.0×10^{-4}	–
BLADE-qSNAP (PyTorch)	CPU fp64, 16 threads	1.9×10^{-4}	8.2×10^{-5}	–

and Surface groups, so meV/atom offsets between crystal structures carry almost no gradient. The fixed bispectrum basis does not have to learn them, and 1595 coefficients resolve them once enough configurations constrain the fit. A loss normalised by each group's energy spread would target these groups directly; we have not tested it.

Full-data forces. At the full-data endpoint BLADE's training force error is 0.0364 against 0.0150 eV/Å for quadratic SNAP, and the Liquid configurations account for the largest share (0.157 eV/Å on Liquid). This is the one headline metric on which BLADE does not reach parity. It is a training error, and on held-out data the order is reversed at every fraction (Sec. 5.1). The ablations show that the gap is not a property of the energy form. $K' = 8$, $K' = 32$ and joint training reach 0.0204, 0.0253 and 0.0258 eV/Å at full data, closer to quadratic SNAP; $K' = 8$ predicts held-out forces as well as BLADE (within 7 %), and the other two worse. The cross-validated objective deliberately does not reward in-sample force accuracy beyond what transfers to held-out configurations.

6.3 Cost

Inference. Table 8 compares MD cost at matched system size. On one CPU core, LAMMPS quadratic SNAP costs 3.0×10^{-4} s per atom-step at every size, and 16 MPI ranks bring this to 2.7×10^{-5} (128 atoms) and 1.9×10^{-5} (8192 atoms). The two descriptors do comparable work per neighbour. BLADE evaluates 8 Chebyshev polynomials and 49 real spherical harmonics (392 real products and their gradients). SNAP evaluates 285 complex hyperspherical-harmonic coefficients for $2J = 8$. Per atom, BLADE then forms 666 invariants and evaluates a $666 \to 128 \to 128 \to 16$ network, while SNAP forms 55 bispectrum components through the full Clebsch–Gordan coupling. Our PyTorch implementation has not been optimised for MD. On one CPU thread it costs 2.4×10^{-4} s per atom-step for 128 atoms, slightly less than LAMMPS SNAP on one core, but its threads scale poorly. On 16 threads BLADE is 6.9 times slower than LAMMPS on 16 MPI ranks at 128 atoms and 4.1 times slower at 1024 atoms. On one GPU in double precision it costs 1.4×10^{-4} s per atom-step for 128 atoms, where kernel-launch latency dominates. It reaches 1.8×10^{-5} at 1024 atoms, about the cost of SNAP on 16 cores, and 3.1×10^{-6} at 8192 atoms, 6.0 times faster than SNAP on 16 cores. Single precision changes little (2.8×10^{-6} at 8192 atoms). At 128 atoms, the size of the main comparison, the learned descriptor is therefore 6.9 (CPU) or 5.1 (GPU) times more expensive than quadratic SNAP (Table 2). The advantage appears only for systems of thousands of atoms per GPU.

Energy conservation. Both potentials give stable, energy-conserving NVE dynamics with the same integrator settings (128 atoms, 1 fs, 10 ps; `code/timing/md_nve.py`). At a mean temperature of about 300 K, the total-energy drift is 3.0×10^{-6} meV/atom/ps for BLADE in double precision, 1.5×10^{-4} in single precision and 2.7×10^{-5} for LAMMPS quadratic SNAP. At about 2000 K it is 8.2×10^{-5}, 3.7×10^{-4} and 7.3×10^{-5} meV/atom/ps. Fluctuations about the trend are a few μeV/atom in every case (RMS 5.0×10^{-4}–3.7×10^{-3} meV/atom). The learned descriptor is therefore as smooth as the bispectrum on these trajectories, and single precision costs nothing visible over 10 ps.

Training. Quadratic SNAP is essentially free to refit. The bispectrum design matrix of Ta takes 9 s of LAMMPS time once, and each of the 191 draws is a 0.8 s least-squares solve. For the transfer metals these become 92–164 s and 6.1–10.3 s. BLADE trains a descriptor for every draw. 1,500 steps took on average 157 s per Ta draw (319 s for the full set) with two trainings sharing one GPU, or 4.2 GPU-hours for the complete 191-draw Ta sweep. With 2048-atom episodes and two or three trainings sharing the GPU, a transfer-metal draw took 215–308 s on average. The episode's design matrix dominates, together with its derivative with respect to the descriptor. Its cost scales with the number of neighbour pairs, K' and the width of the primitive basis. This cost is paid once per model, but per draw it is about two hundred times SNAP's least-squares solve on Ta, and twenty to fifty times on the transfer metals.

Cost–quality trade-off. For production MD at fixed accuracy on well-sampled near-crystalline configurations, quadratic SNAP in LAMMPS remains the most economical choice. The learned descriptor pays off when training data are scarce or the configurations of interest are disordered or require extrapolation in density. There it reaches errors quadratic SNAP cannot reach at any training fraction in our sweeps. The price is a one-off training cost of a few GPU-minutes per model and an inference cost that, in our unoptimised implementation, exceeds SNAP's for small systems and falls below it from about a thousand atoms per GPU. With abundant data, as for the transfer metals, quadratic SNAP is both cheaper and, on Nb and Mo, more accurate.

7 Related work

SNAP and its descendants. SNAP [1] and quadratic SNAP [4] fit linear models in fixed bispectrum components. They have been extended to multiple elements [15] and deployed for refractory metals and alloys [16–19]. Their hyper-parameters (cutoff, group weights) are tuned by a gradient-free outer loop around the closed-form fit (DAKOTA), which is a coarse, black-box version of the bilevel problem we solve with gradients. Neural extensions keep the bispectrum fixed and make the *head* nonlinear [20–23]. FitSNAP also offers PyTorch networks on fixed SNAP, ACE or pairwise descriptors [13]. BLADE does the opposite: it learns the descriptor and keeps the convex, closed-form head of Eq. (1).

Learned atom-centred descriptors. The density-expansion building blocks of BLADE have long been in use. These are radial channels formed as learned combinations of a fixed radial basis, angular expansion in spherical harmonics, and contraction to power-spectrum and bispectrum invariants. MTP [3] and ACE/PACE [2, 24–26] fit radial coefficients together with a linear or embedded energy. NEP [7, 27] uses Chebyshev radial functions with trainable mixing and spherical-harmonic invariants feeding a small network, which is the closest descriptor *form* to ours. Gaussian moments [28], EANN [29] and the embedding network of DeepPot-SE [30] learn radial channels in related ways. Learned compression of fixed descriptors has been studied in several works [31–35]. In particular, Willatt et al. learn a linear mixing of SOAP species channels for a ridge model [31]. Message-passing networks such as SchNet, PhysNet, PaiNN, NequIP, Allegro and MACE [36–41] learn the representation end to end with a nonlinear readout, and so do not separate descriptor and head. We claim no novelty for the descriptor form. Our contribution is the way the descriptor is trained and the regression form it is trained for.

Closed-form heads, bilevel and episodic training. Differentiating through a closed-form ridge or convex head inside support/query episodes is established in few-shot classification (R2-D2, MetaOptNet [9, 10]). Related techniques include deep kernels and Bayesian last layers [42–45], and its chemistry analogue, ADKF-IFT [46], predicts molecular properties without forces. Eliminating linear parameters in closed form is the classical variable-projection method [11, 47–49]. For MLIPs, two lines of work come closest. Schmitz et al. differentiate through a kernel solve to tune two hyper-parameters [50]. Frozen foundation-model features with closed-form ridge heads [51–54] never optimise the features through the solve. Meta-learning for potentials has so far been applied across levels of theory with Reptile [55]. To our knowledge, no prior potential trains its atomic descriptor through the least-squares solution of its own linear-in-parameters energy head, nor uses within-material support/query episodes to target the small-data regime where quadratic SNAP is known to overfit.

Symmetry: built in or learned. Rotation equivariance can be enforced architecturally, symmetrised a posteriori [56], or learned approximately from data or augmentation [57, 58]. BLADE builds all three invariances in. Its non-invariant ablation in Sec. 5.4 quantifies what is lost when rotation invariance has to be learned. Body-ordered power spectra and bispectra are known to be incomplete [59, 60]. The nonlinear compression in BLADE mixes them but cannot, in principle, recover information they do not contain.

Data efficiency of interatomic potentials. Learning curves of SNAP, quadratic SNAP, GAP, MTP and neural potentials on refractory metals show that quadratic SNAP gains most from additional data and overfits at large $J_{\max}$ [61]. Systematic data-efficiency studies emphasise representation and training-set choice [62, 63]. Our CV sweeps follow the protocol of the original quadratic-SNAP paper exactly, so that the two descriptors can be compared draw by draw.

8 Conclusions and limitations

We replaced the bispectrum of quadratic SNAP by a learned descriptor and held everything else fixed: the energy form, the isolated-atom constraint, analytic forces and the least-squares solve. The descriptor was trained through the closed-form solution of its own head to minimise that head's cross-validated error. Whether the substitution pays off depends on the data regime. On Wood and Thompson's tantalum set, where quadratic SNAP's 1595 coefficients are underdetermined at small fractions, the learned descriptor removes the overfitting. Its equal-weight and pooled held-out errors are lower at all 19 fractions, by orders of magnitude below 30 % of the data. Energy parity holds at full data for each of five training seeds and on the 80/20 split, and the GSF, Liquid and Surface subset reaches quadratic SNAP's full-data energy error with 0.55 of the data. On the transfer metals, whose datasets are an order of magnitude larger, quadratic SNAP is overdetermined at every fraction, and the advantage narrows to the smallest fractions and to particular configuration classes. At the smallest fractions BLADE is still the more accurate model on all three metals, by factors of 1.48 to 2.14 in the equal-weight energy at $f = 0.05$, and on Nb it predicts defect energies better at every fraction. In the data-rich regime it is at parity on Re, slightly behind on Nb and behind on Mo, and its pooled force errors are higher on all three. On every transfer metal BLADE's error stops decreasing with data, while its training error stays close to its held-out error. With thousands of configurations, the learned descriptor is limited by how well it can fit, not by overfitting, and quadratic SNAP's larger head overtakes it. In double precision the learned descriptor is invariant to round-off on all four metals, and in molecular dynamics it conserves energy as well as LAMMPS SNAP.

The ablations locate the gains. Training the same descriptor on its in-sample error instead of the out-of-fold error of its head doubles the held-out energy error. Without group balancing, the subset energy errors are two and a half to four times higher once $f \geq 0.3$. Without the invariant contraction, held-out forces are two to three and a half times worse even though no test configuration is ever rotated, and rotation augmentation does not recover the loss within the same training budget. A linear compression of the bispectrum trained with the same objective recovers most of the force generalisation but little of the energy accuracy, which therefore lives in the nonlinear part of the learned features. The failures are equally specific. On near-crystalline configurations with abundant data, on perfect and elastically deformed crystals of Nb, and in full-data forces, quadratic SNAP is more accurate. There, errors are meV/atom offsets between crystal structures and fine force detail, which a fixed, fine basis with many coefficients resolves better than a descriptor trained under a loss dominated by large errors.

Limitations. Several limitations qualify these conclusions. The comparison concerns single-element potentials at one bispectrum order, $J_{\max} = 4$. A learned descriptor for alloys needs species channels, and at $J_{\max} = 5$ quadratic SNAP has 4277 coefficients, which shifts the balance between coefficient count and data again. BLADE's hyper-parameters were frozen on Ta development splits and used unchanged for the transfer metals. This strengthens the transfer test, but the ablations show that the frozen choice is not optimal: $K' = 8$ predicts Ta about as well as the frozen $K' = 16$ and fits it better. The transfer sweeps use smaller training episodes than Ta for compute reasons, which on Re cost 13 % in held-out energy accuracy (Sec. 5.3). Training is stochastic and not bitwise reproducible, so single runs can differ substantially; our conclusions rest on ten-draw means and on the five-seed spread of the full-data fits. Training costs one to two orders of magnitude more than refitting SNAP, and our molecular-dynamics implementation is a PyTorch prototype rather than a LAMMPS pair

style. The data-efficiency statistic $F^\star$ requires a sustained crossing of a training-error reference. It is ill-suited to flat learning curves such as BLADE's on Re, and, as Nb shows, the reference itself can be soft. Finally, our reimplementation reproduces the published quadratic model but not the published linear one, and the provided Ta data differ from the paper's description in three groups.

Outlook. Three directions follow directly from the failure analysis. A group loss normalised by each group's energy spread would direct the descriptor's training toward the meV/atom differences between crystal structures that it currently neglects. Training episodes that hold more configurations than the head has coefficients should recover energy accuracy on the transfer metals. On Re they lowered the held-out energy error at $f = 0.5$ by 13 %; whether they close the gap on Nb and Mo is the most direct open question. And because the descriptor is trained only through a linear-in-parameters head, the same objective could be used to adapt the radial functions of ACE or MTP while keeping their fitting pipelines unchanged.

References

[1] A.P. Thompson, L.P. Swiler, C.R. Trott, S.M. Foiles, and G.J. Tucker. Spectral neighbor analysis method for automated generation of quantum-accurate interatomic potentials. *Journal of Computational Physics*, 285:316–330, 2015. doi: 10.1016/j.jcp.2014.12.018.

[2] Ralf Drautz. Atomic cluster expansion for accurate and transferable interatomic potentials. *Physical Review B*, 99(1):014104, 2019. doi: 10.1103/PhysRevB.99.014104.

[3] Alexander V. Shapeev. Moment Tensor Potentials: A Class of Systematically Improvable Interatomic Potentials. *Multiscale Modeling & Simulation*, 14(3):1153–1173, 2016. doi: 10.1137/15M1054183.

[4] Mitchell A. Wood and Aidan P. Thompson. Extending the accuracy of the SNAP interatomic potential form. *The Journal of Chemical Physics*, 148(24):241721, 2018. doi: 10.1063/1.5017641.

[5] Christopher M. Andolina and Wissam A. Saidi. Highly transferable atomistic machine-learning potentials from curated and compact datasets across the periodic table. *Digital Discovery*, 2(4):1070–1077, 2023. doi: 10.1039/D3DD00046J.

[6] Albert P. Bartók, Risi Kondor, and Gábor Csányi. On representing chemical environments. *Physical Review B*, 87(18):184115, 2013. doi: 10.1103/PhysRevB.87.184115.

[7] Zheyong Fan, Zezhu Zeng, Cunzhi Zhang, Yanzhou Wang, Keke Song, Haikuan Dong, Yue Chen, and Tapio Ala-Nissila. Neuroevolution machine learning potentials: Combining high accuracy and low cost in atomistic simulations and application to heat transport. *Physical Review B*, 104(10):104309, 2021. doi: 10.1103/PhysRevB.104.104309.

[8] Michael J. Willatt, Félix Musil, and Michele Ceriotti. Atom-density representations for machine learning. *The Journal of Chemical Physics*, 150(15):154110, 2019. doi: 10.1063/1.5090481.

[9] Luca Bertinetto, João F. Henriques, Philip H. S. Torr, and Andrea Vedaldi. Meta-learning with differentiable closed-form solvers. In *International Conference on Learning Representations (ICLR)*, 2019. URL https://openreview.net/forum?id=HyxnZh0ct7.

[10] Kwonjoon Lee, Subhransu Maji, Avinash Ravichandran, and Stefano Soatto. Meta-Learning With Differentiable Convex Optimization. In *2019 IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pages 10649–10657, 2019. doi: 10.1109/CVPR.2019.01091.

[11] G. H. Golub and V. Pereyra. The Differentiation of Pseudo-Inverses and Nonlinear Least Squares Problems Whose Variables Separate. *SIAM Journal on Numerical Analysis*, 10(2):413–432, 1973. doi: 10.1137/0710036.

[12] Saidi Group. 23-Single-Element-DNPs: DFT datasets for training machine learning atomistic potentials, final DNP version, and example LAMMPS/VASP validation scripts. https://github.com/saidigroup/23-Single-Element-DNPs, 2023. GitHub repository, GPL-3.0; created 2023-03-15, last push 2024-01-09; accessed 2026-10-03. Companion data of Andolina and Saidi (2023).

[13] A. Rohskopf, C. Sievers, N. Lubbers, M. A. Cusentino, J. Goff, J. Janssen, M. McCarthy, D. Montes de Oca Zapiain, S. Nikolov, K. Sargsyan, D. Sema, E. Sikorski, L. Williams, A. P. Thompson, and M. A. Wood. FitSNAP: Atomistic machine learning with LAMMPS. *Journal of Open Source Software*, 8(84):5118, 2023. doi: 10.21105/joss.05118.

[14] Aidan P. Thompson, H. Metin Aktulga, Richard Berger, Dan S. Bolintineanu, W. Michael Brown, Paul S. Crozier, Pieter J. in 't Veld, Axel Kohlmeyer, Stan G. Moore, Trung Dac Nguyen, Ray Shan, Mark J. Stevens, Julien Tranchida, Christian Trott, and Steven J. Plimpton. LAMMPS - a flexible simulation tool for particle-based materials modeling at the atomic, meso, and continuum scales. *Computer Physics Communications*, 271:108171, 2022. doi: 10.1016/j.cpc.2021.108171.

[15] M. A. Cusentino, M. A. Wood, and A. P. Thompson. Explicit Multielement Extension of the Spectral Neighbor Analysis Potential for Chemically Complex Systems. *The Journal of Physical Chemistry A*, 124(26):5456–5464, 2020. doi: 10.1021/acs.jpca.0c02450.

[16] Mitchell A. Wood and Aidan P. Thompson. Quantum-Accurate Molecular Dynamics Potential for Tungsten, 2017.

[17] M. A. Wood, M. A. Cusentino, B. D. Wirth, and A. P. Thompson. Data-driven material models for atomistic simulation. *Physical Review B*, 99(18):184305, 2019. doi: 10.1103/PhysRevB.99.184305.

[18] Chi Chen, Zhi Deng, Richard Tran, Hanmei Tang, Iek-Heng Chu, and Shyue Ping Ong. Accurate force field for molybdenum by machine learning large materials data. *Physical Review Materials*, 1(4):043603, 2017. doi: 10.1103/PhysRevMaterials.1.043603.

[19] Xiang-Guo Li, Chongze Hu, Chi Chen, Zhi Deng, Jian Luo, and Shyue Ping Ong. Quantum-accurate spectral neighbor analysis potential models for Ni-Mo binary alloys and fcc metals. *Physical Review B*, 98(9):094104, 2018. doi: 10.1103/PhysRevB.98.094104.

[20] Howard Yanxon, David Zagaceta, Brandon C. Wood, and Qiang Zhu. Neural network potential from bispectrum components: A case study on crystalline silicon. *The Journal of Chemical Physics*, 153(5):054118, 2020. doi: 10.1063/5.0014677.

[21] David Zagaceta, Howard Yanxon, and Qiang Zhu. Spectral neural network potentials for binary alloys. *Journal of Applied Physics*, 128(4):045113, 2020. doi: 10.1063/5.0013208.

[22] David Montes de Oca Zapiain, Mitchell A. Wood, Nicholas Lubbers, Carlos Z. Pereyra, Aidan P. Thompson, and Danny Perez. Training data selection for accuracy and transferability of interatomic potentials. *npj Computational Materials*, 8(1):189, 2022. doi: 10.1038/s41524-022-00872-x.

[23] A. Rohskopf, J. Goff, D. Sema, K. Gordiz, N. C. Nguyen, A. Henry, A. P. Thompson, and M. A. Wood. Exploring model complexity in machine learned potentials for simulated properties. *Journal of Materials Research*, 38(24):5136–5150, 2023. doi: 10.1557/s43578-023-01152-0.

[24] Yury Lysogorskiy, Cas van der Oord, Anton Bochkarev, Sarath Menon, Matteo Rinaldi, Thomas Hammerschmidt, Matous Mrovec, Aidan Thompson, Gábor Csányi, Christoph Ortner, and Ralf Drautz. Performant implementation of the atomic cluster expansion (PACE) and application to copper and silicon. *npj Computational Materials*, 7(1):97, 2021. doi: 10.1038/s41524-021-00559-9.

[25] Anton Bochkarev, Yury Lysogorskiy, Sarath Menon, Minaam Qamar, Matous Mrovec, and Ralf Drautz. Efficient parametrization of the atomic cluster expansion. *Physical Review Materials*, 6 (1):013804, 2022. doi: 10.1103/PhysRevMaterials.6.013804.

[26] Geneviève Dusson, Markus Bachmayr, Gábor Csányi, Ralf Drautz, Simon Etter, Cas van der Oord, and Christoph Ortner. Atomic cluster expansion: Completeness, efficiency and stability. *Journal of Computational Physics*, 454:110946, 2022. doi: 10.1016/j.jcp.2022.110946.

[27] Zheyong Fan. Improving the accuracy of the neuroevolution machine learning potential for multi-component systems. *Journal of Physics: Condensed Matter*, 34(12):125902, 2022. doi: 10.1088/1361-648X/ac462b.

[28] V. Zaverkin and J. Kästner. Gaussian Moments as Physically Inspired Molecular Descriptors for Accurate and Scalable Machine Learning Potentials. *Journal of Chemical Theory and Computation*, 16(8):5410–5421, 2020. doi: 10.1021/acs.jctc.0c00347.

[29] Yaolong Zhang, Ce Hu, and Bin Jiang. Embedded Atom Neural Network Potentials: Efficient and Accurate Machine Learning with a Physically Inspired Representation. *The Journal of Physical Chemistry Letters*, 10(17):4962–4967, 2019. doi: 10.1021/acs.jpclett.9b02037.

[30] Linfeng Zhang, Jiequn Han, Han Wang, Wissam A. Saidi, Roberto Car, and Weinan E. End-to-end Symmetry Preserving Inter-atomic Potential Energy Model for Finite and Extended Systems. In *Advances in Neural Information Processing Systems*, volume 31. Curran Associates, Inc., 2018. URL https://proceedings.neurips.cc/paper_files/paper/2018/hash/e2ad76f2326fbc6b56a45a56c59fafdb-Abstract.html.

[31] Michael J. Willatt, Félix Musil, and Michele Ceriotti. Feature optimization for atomistic machine learning yields a data-driven construction of the periodic table of the elements. *Physical Chemistry Chemical Physics*, 20(47):29661–29668, 2018. doi: 10.1039/C8CP05921G.

[32] Alexander Goscinski, Félix Musil, Sergey Pozdnyakov, Jigyasa Nigam, and Michele Ceriotti. Optimal radial basis for density-based atomic representations. *The Journal of Chemical Physics*, 155(10):104106, 2021. doi: 10.1063/5.0057229.

[33] James P. Darby, James R. Kermode, and Gábor Csányi. Compressing local atomic neighbourhood descriptors. *npj Computational Materials*, 8(1):166, 2022. doi: 10.1038/s41524-022-00847-y.

[34] James P. Darby, Dávid P. Kovács, Ilyes Batatia, Miguel A. Caro, Gus L. W. Hart, Christoph Ortner, and Gábor Csányi. Tensor-Reduced Atomic Density Representations. *Physical Review Letters*, 131(2):028001, 2023. doi: 10.1103/PhysRevLett.131.028001.

[35] Claudio Zeni, Kevin Rossi, Aldo Glielmo, and Stefano de Gironcoli. Compact atomic descriptors enable accurate predictions via linear models. *The Journal of Chemical Physics*, 154(22):224112, 2021. doi: 10.1063/5.0052961.

[36] K. T. Schütt, H. E. Sauceda, P.-J. Kindermans, A. Tkatchenko, and K.-R. Müller. SchNet – A deep learning architecture for molecules and materials. *The Journal of Chemical Physics*, 148 (24):241722, 2018. doi: 10.1063/1.5019779.

[37] Oliver T. Unke and Markus Meuwly. PhysNet: A Neural Network for Predicting Energies, Forces, Dipole Moments, and Partial Charges. *Journal of Chemical Theory and Computation*, 15(6):3678–3693, 2019. doi: 10.1021/acs.jctc.9b00181.

[38] Kristof Schütt, Oliver Unke, and Michael Gastegger. Equivariant message passing for the prediction of tensorial properties and molecular spectra. In *Proceedings of the 38th International Conference on Machine Learning*, volume 139 of *Proceedings of Machine Learning Research*, pages 9377–9388. PMLR, 2021. URL https://proceedings.mlr.press/v139/schutt21a.html.

[39] Simon Batzner, Albert Musaelian, Lixin Sun, Mario Geiger, Jonathan P. Mailoa, Mordechai Kornbluth, Nicola Molinari, Tess E. Smidt, and Boris Kozinsky. E(3)-equivariant graph neural networks for data-efficient and accurate interatomic potentials. *Nature Communications*, 13(1): 2453, 2022. doi: 10.1038/s41467-022-29939-5.

[40] Albert Musaelian, Simon Batzner, Anders Johansson, Lixin Sun, Cameron J. Owen, Mordechai Kornbluth, and Boris Kozinsky. Learning local equivariant representations for large-scale atomistic dynamics. *Nature Communications*, 14(1):579, 2023. doi: 10.1038/s41467-023-36329-y.

[41] Ilyes Batatia, Dávid Péter Kovács, Gregor N. C. Simm, Christoph Ortner, and Gábor Csányi. MACE: Higher Order Equivariant Message Passing Neural Networks for Fast and Accurate Force Fields. In *Advances in Neural Information Processing Systems*, volume 35, pages 11423–11436. Curran Associates, Inc., 2022. doi: 10.52202/068431-0830.

[42] Jasper Snoek, Oren Rippel, Kevin Swersky, Ryan Kiros, Nadathur Satish, Narayanan Sundaram, Mostofa Patwary, Prabhat, and Ryan Adams. Scalable Bayesian Optimization Using Deep Neural Networks. In *Proceedings of the 32nd International Conference on Machine Learning*, volume 37 of *Proceedings of Machine Learning Research*, pages 2171–2180. PMLR, 2015. URL https://proceedings.mlr.press/v37/snoek15.html.

[43] Andrew Gordon Wilson, Zhiting Hu, Ruslan Salakhutdinov, and Eric P. Xing. Deep Kernel Learning. In *Proceedings of the 19th International Conference on Artificial Intelligence and Statistics*, volume 51 of *Proceedings of Machine Learning Research*, pages 370–378. PMLR, 2016. URL https://proceedings.mlr.press/v51/wilson16.html.

[44] James Harrison, Apoorva Sharma, and Marco Pavone. Meta-learning Priors for Efficient Online Bayesian Regression. In *Algorithmic Foundations of Robotics XIII*, Springer Proceedings in Advanced Robotics, pages 318–337. Springer International Publishing, 2020. doi: 10.1007/978-3-030-44051-0_19.

[45] Massimiliano Patacchiola, Jack Turner, Elliot J. Crowley, Michael O'Boyle, and Amos Storkey. Bayesian Meta-Learning for the Few-Shot Setting via Deep Kernels. In *Advances in Neural Information Processing Systems*, volume 33, pages 16108–16118. Curran Associates, Inc., 2020. URL https://proceedings.neurips.cc/paper_files/paper/2020/hash/b9cfe8b6042cf759dc4c0cccb27a6737-Abstract.html.

[46] Wenlin Chen, Austin Tripp, and José Miguel Hernández-Lobato. Meta-learning Adaptive Deep Kernel Gaussian Processes for Molecular Property Prediction. In *International Conference on Learning Representations (ICLR)*, 2023.

[47] Gene Golub and Victor Pereyra. Separable nonlinear least squares: the variable projection method and its applications. *Inverse Problems*, 19(2):R1–R26, 2003. doi: 10.1088/0266-5611/19/2/201.

[48] Elizabeth Newman, Lars Ruthotto, Joseph Hart, and Bart van Bloemen Waanders. Train Like a (Var)Pro: Efficient Training of Neural Networks with Variable Projection. *SIAM Journal on Mathematics of Data Science*, 3(4):1041–1066, 2021. doi: 10.1137/20M1359511.

[49] Eric C. Cyr, Mamikon A. Gulian, Ravi G. Patel, Mauro Perego, and Nathaniel A. Trask. Robust Training and Initialization of Deep Neural Networks: An Adaptive Basis Viewpoint. In *Proceedings of The First Mathematical and Scientific Machine Learning Conference*, volume 107 of *Proceedings of Machine Learning Research*, pages 512–536. PMLR, 2020. URL https://proceedings.mlr.press/v107/cyr20a.html.

[50] Niklas Frederik Schmitz, Klaus-Robert Müller, and Stefan Chmiela. Algorithmic Differentiation for Automated Modeling of Machine Learned Force Fields. *The Journal of Physical Chemistry Letters*, 13(43):10183–10189, 2022. doi: 10.1021/acs.jpclett.2c02632.

[51] Pietro Novelli, Giacomo Meanti, Pedro J. Buigues, Lorenzo Rosasco, Michele Parrinello, Massimiliano Pontil, and Luigi Bonati. Fast and Fourier features for transfer learning of interatomic potentials. *npj Computational Materials*, 11(1):293, 2025. doi: 10.1038/s41524-025-01779-z.

[52] John Falk, Luigi Bonati, Pietro Novelli, Michele Parrinello, and Massimiliano Pontil. Transfer learning for atomistic simulations using GNNs and kernel mean embeddings. In *Advances in Neural Information Processing Systems*, volume 36, pages 29783–29797. Curran Associates, Inc., 2023. doi: 10.52202/075280-1296.

[53] Filippo Bigi, Sanggyu Chong, Michele Ceriotti, and Federico Grasselli. A prediction rigidity formalism for low-cost uncertainties in trained neural networks. *Machine Learning: Science and Technology*, 5(4):045018, 2024. doi: 10.1088/2632-2153/ad805f.

[54] Mariia Radova, Wojciech G. Stark, Connor S. Allen, Reinhard J. Maurer, and Albert P. Bartók. Fine-tuning foundation models of materials interatomic potentials with frozen transfer learning. *npj Computational Materials*, 11(1):237, 2025. doi: 10.1038/s41524-025-01727-x.

[55] Alice E. A. Allen, Nicholas Lubbers, Sakib Matin, Justin Smith, Richard Messerly, Sergei Tretiak, and Kipton Barros. Learning together: Towards foundation models for machine learning interatomic potentials with meta-learning. *npj Computational Materials*, 10(1):154, 2024. doi: 10.1038/s41524-024-01339-x.

[56] Sergey N. Pozdnyakov and Michele Ceriotti. Smooth, exact rotational symmetrization for deep learning on point clouds. In *Advances in Neural Information Processing Systems*, volume 36, pages 79469–79501. Curran Associates, Inc., 2023. doi: 10.52202/075280-3478.

[57] Marcel F Langer, Sergey N Pozdnyakov, and Michele Ceriotti. Probing the effects of broken symmetries in machine learning. *Machine Learning: Science and Technology*, 5(4):04LT01, 2024. doi: 10.1088/2632-2153/ad86a0.

[58] Eric Qu and Aditi S. Krishnapriyan. The Importance of Being Scalable: Improving the Speed and Accuracy of Neural Network Interatomic Potentials Across Chemical Domains. In *Advances in Neural Information Processing Systems*, volume 37, pages 139030–139053. Curran Associates, Inc., 2024. doi: 10.52202/079017-4412.

[59] Sergey N. Pozdnyakov, Michael J. Willatt, Albert P. Bartók, Christoph Ortner, Gábor Csányi, and Michele Ceriotti. Incompleteness of Atomic Structure Representations. *Physical Review Letters*, 125(16):166001, 2020. doi: 10.1103/PhysRevLett.125.166001.

[60] Jigyasa Nigam, Sergey Pozdnyakov, and Michele Ceriotti. Recursive evaluation and iterative contraction of N-body equivariant features. *The Journal of Chemical Physics*, 153(12):121101, 2020. doi: 10.1063/5.0021116.

[61] Yunxing Zuo, Chi Chen, Xiangguo Li, Zhi Deng, Yiming Chen, Jörg Behler, Gábor Csányi, Alexander V. Shapeev, Aidan P. Thompson, Mitchell A. Wood, and Shyue Ping Ong. Performance and Cost Assessment of Machine Learning Interatomic Potentials. *The Journal of Physical Chemistry A*, 124(4):731–745, 2020. doi: 10.1021/acs.jpca.9b08723.

[62] Joshua A Vita and Daniel Schwalbe-Koda. Data efficiency and extrapolation trends in neural network interatomic potentials. *Machine Learning: Science and Technology*, 4(3):035031, 2023. doi: 10.1088/2632-2153/acf115.

[63] Marcel F. Langer, Alex Goeßmann, and Matthias Rupp. Representations of molecules and materials for interpolation of quantum-mechanical simulations via machine learning. *npj Computational Materials*, 8(1):41, 2022. doi: 10.1038/s41524-022-00721-x.

A Development record and leakage control

Table 9 lists every development run made before the cross-validation sweeps, in chronological order. The source is `logs/dev/dev_runs.jsonl`; the narrative is in `attempts_log.md`. Three splits were used:

- *dev_full*: the training error of a fit to all 363 configurations, which is the paper's own full-data endpoint;
- *dev_half*: an 80/20 split (148/37) of the configurations *included* in sweep draw 90 ($f = 0.50$, draw 0);
- *dev_small*: the same for draw 10 ($f = 0.10$, draw 0; 30 training configurations).

Validation configurations of these splits belong to the held-out sets of other sweep draws, which is unavoidable on a fixed dataset; the published r_{cut} of quadratic SNAP was likewise chosen on all 363 configurations [4]. No held-out configuration of a sweep draw was used for a decision about that draw.

One episode needs stating explicitly. A first sweep with the v1 configuration was stopped after six completed draws. The full-data draw's per-group *training* errors showed that v1 underfits the three Liquid configurations, a defect that was already present in the dev runs' training errors but hidden by pooled metrics. Those six draws were discarded (`logs/cv_discarded/`). They included five held-out results at $f = 0.95$, which we had seen; the subsequent changes were selected on dev splits and training errors only.

Table 9: Development runs on Ta (all errors in eV/atom and eV/Å). Full data: training errors; dev splits: validation errors. "v0"/"v1": earlier frozen candidates; bold: the configuration used in all sweeps.

	full data (training)			dev_half (val.)			dev_small (val.)	
run	δE	δF	Liq. δE	E_{eqw}	E_{sub}	δF	E_{eqw}	δF
quadratic SNAP (reference)	0.0082	0.015	0.0087	10.205	0.030	0.433	2.3e4	2.761
v0: random split episodes, $\ell \leq 4$, 8 ch., lr 2×10^{-3}, 3000 st.	0.0088	0.033	0.178	0.095	0.089	0.129	1.140	0.015
v0, lr 5×10^{-3}	0.011	0.036	0.083	0.050	0.016	0.101	0.915	0.018
v0, lr 10^{-3}	0.012	0.035	0.329	0.113	0.098	0.175	1.406	0.015
v0, $w_E = 10$	0.0033	0.035	0.056	0.061	0.035	0.130	0.942	0.012
v0, $K' = 32$	0.016	0.026	0.426	0.073	0.077	0.161	0.842	0.013
joint (SGD head), 3000 st.	0.0046	0.022	0.051	0.042	0.0025	0.301	2.309	0.015
stratified split episodes, 2000 st., 128-config episodes	0.017	0.045	0.123	–	–	–	–	–
as above, $K' = 24$	0.018	0.038	0.083	–	–	–	–	–
as above, $\ell \leq 6$	0.024	0.047	0.136	–	–	–	–	–
joint, 2000 st., 128-config episodes	0.011	0.031	0.110	–	–	–	–	–
k-fold, singletons in support (v1 base)	0.0054	0.028	0.149	0.091	0.094	0.134	2.158	0.109
k-fold, random folds	0.0077	0.038	0.192	0.089	0.062	0.123	1.507	0.012
k-fold, singletons in support, $K' = 24$	0.0090	0.029	0.299	0.114	0.093	0.198	1.642	0.055
joint, 1500 st.	0.011	0.030	0.130	0.056	0.018	0.304	2.003	0.020
k-fold (v1), $\ell \leq 6$	0.0050	0.028	0.117	0.069	0.049	0.121	0.736	0.018
k-fold (v1), $\ell \leq 6$, 12 ch., MLP 128	0.0040	0.028	0.109	0.053	0.061	0.102	1.279	0.012
k-fold (v1), 12 radial, 12 ch., MLP 128	0.0035	0.024	0.115	0.087	0.070	0.109	2.924	0.018
k-fold (v1), $\ell \leq 6$, $K' = 24$	0.0066	0.028	0.144	0.064	0.085	0.131	1.711	0.019
group-balanced, base	0.0069	0.035	0.024	0.074	0.0090	0.141	0.569	0.017
group-balanced, $\ell \leq 4$ wide	0.0029	0.027	0.0079	0.064	0.0042	0.140	1.665	0.015
group-balanced, $\ell \leq 6$ wide (frozen v2)	0.0044	0.035	0.014	0.017	0.0036	0.097	0.637	0.014
$w_E = 10$, $\ell \leq 4$ wide	0.0023	0.037	0.035	0.046	0.016	0.114	2.519	0.019
group-balanced $\ell \leq 4$ wide, 2048-atom episodes	0.0094	0.037	0.048	–	–	–	–	–

B Per-group results

Table 10: Ta: per-group mean held-out energy error (eV/atom) at selected fractions and full-data training error, quadratic SNAP / BLADE-qSNAP. Means over 10 draws; standard deviations for all fractions are in `logs/cv/`.

group	f=0.10	f=0.30	f=0.50	f=0.80	full data
Displaced_A15	1.68 / 0.0387	0.0035 / 0.0090	0.0070 / 0.0099	0.0086 / 0.0030	0.0314 / 0.0112
Displaced_BCC	5.57 / 0.0125	0.0056 / 0.0057	0.0023 / 0.0058	0.0010 / 0.0029	0.0047 / 0.0030
Displaced_FCC	0.350 / 0.0091	0.0006 / 0.0055	0.0006 / 0.0036	0.0006 / 0.0012	0.0006 / 0.0003
Elastic_BCC	0.0027 / 0.0123	0.0007 / 0.0036	0.0006 / 0.0041	0.0008 / 0.0011	0.0027 / 0.0026
Elastic_FCC	0.0053 / 0.0107	0.0025 / 0.0066	0.0022 / 0.0038	0.0017 / 0.0020	0.0035 / 0.0019
GSF_110	2.82 / 0.0198	0.0010 / 0.0083	0.0009 / 0.0046	0.0015 / 0.0042	0.0080 / 0.0024
GSF_112	6.97 / 0.0118	0.0023 / 0.0071	0.0017 / 0.0063	0.0020 / 0.0031	0.0062 / 0.0027
Liquid	50.0 / 0.112	0.0639 / 0.0430	0.0331 / 0.0240	0.0246 / 0.0120	0.0087 / 0.0165
Surface	85.7 / 0.0417	0.123 / 0.0203	0.111 / 0.0198	0.0149 / 0.0052	0.0227 / 0.0048
Volume_A15	4.2×10^{4} / 0.541	8.24 / 0.126	0.955 / 0.0445	0.449 / 0.0246	0.0155 / 0.0167
Volume_BCC	2.0×10^{6} / 1.62	494 / 0.441	42.9 / 0.146	14.8 / 0.107	0.0147 / 0.0224
Volume_FCC	1.4×10^{5} / 0.831	53.4 / 0.268	3.16 / 0.0685	2.57 / 0.0686	0.0247 / 0.0313

C Per-group results for the transfer metals

Figures 7–12 resolve every group of the transfer metals, and Tables 12–14 list the per-group values at selected fractions. Table 11 splits the held-out energy errors by the energy of the configuration above the lowest energy per atom of the material.

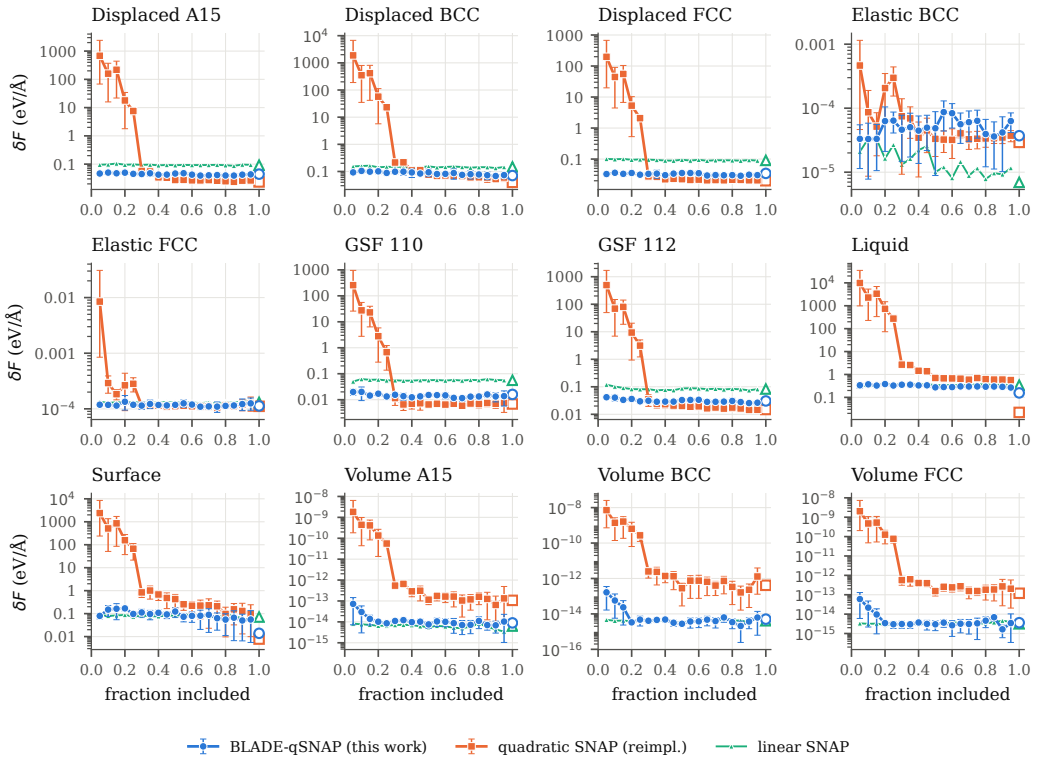


Figure 6: Per-group held-out force errors on Ta (mean ± std over 10 draws; open symbols: full-data training error). The DFT forces of the Volume and Elastic groups vanish by symmetry (below 0.01 eV/Å), so their force errors measure numerical noise and are not interpreted.

Table 11: Transfer metals: held-out energy MAE (eV/atom) by energy above the lowest energy per atom of the material, pooled over the 10 draws at each fraction (`code/analysis/energy_bins.py`). Share: fraction of all configurations in the bin.

	$E/N - \min E/N$	share	$f = 0.05$		$f = 0.50$		$f = 0.80$	
	(eV/atom)	(%)	BLADE	qSNAP	BLADE	qSNAP	BLADE	qSNAP
Mo	0–0.5	29	0.0375	0.0359	0.0465	0.0279	0.0455	0.0255
	0.5–1	27	0.0352	0.0576	0.0419	0.0416	0.0431	0.0409
	1–2	18	0.0436	0.0668	0.0448	0.0427	0.0509	0.0386
	2–5	20	0.0902	0.113	0.0909	0.0919	0.0970	0.0862
	≥ 5	6	0.326	0.570	0.157	0.186	0.135	0.162
Nb	0–0.5	24	0.0202	0.0241	0.0274	0.0175	0.0272	0.0181
	0.5–1	32	0.0230	0.0299	0.0312	0.0231	0.0271	0.0243
	1–2	27	0.0359	0.0521	0.0411	0.0404	0.0439	0.0425
	2–5	13	0.0684	0.107	0.0731	0.0694	0.0711	0.0712
	≥ 5	4	0.570	0.731	0.174	0.183	0.146	0.190
Re	0–0.5	18	0.0122	0.0202	0.0132	0.0169	0.0121	0.0156
	0.5–1	33	0.0171	0.0282	0.0195	0.0185	0.0182	0.0174
	1–2	25	0.0213	0.0489	0.0285	0.0278	0.0263	0.0264
	2–5	18	0.0256	0.0501	0.0264	0.0358	0.0314	0.0334
	≥ 5	6	0.0702	0.214	0.0438	0.0716	0.0362	0.0629

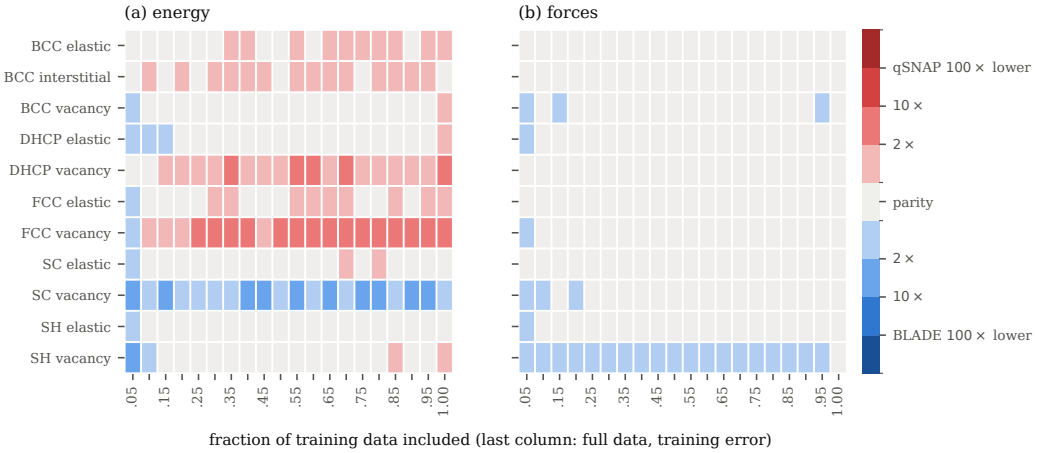


Figure 7: Mo: ratio of 10-draw mean held-out errors, quadratic SNAP over BLADE, per group and included fraction (as Fig. 3). Blue: BLADE lower; red: quadratic SNAP lower; grey: within a factor of 1.26.

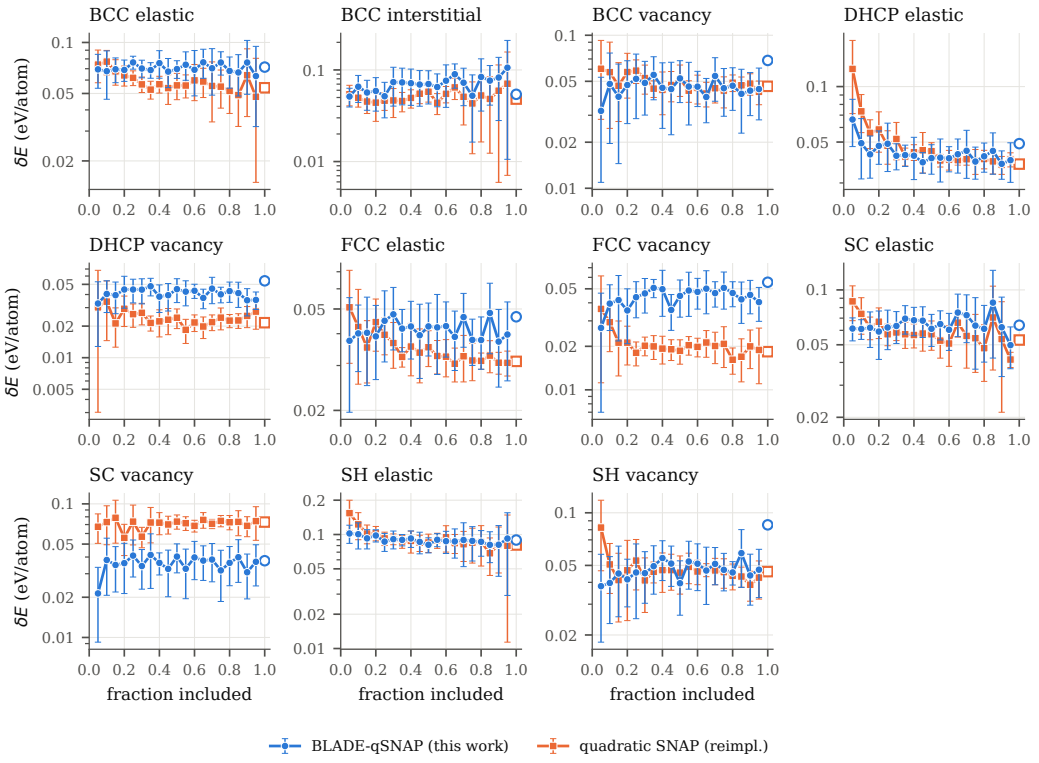


Figure 8: Mo: per-group held-out energy errors (mean ± std over 10 draws; open symbols: full-data training errors).



Figure 9: Re: $\log_{10}$ of the ratio of 10-draw mean held-out errors, quadratic SNAP over BLADE, per group and included fraction (as Fig. 3). Blue: BLADE lower; red: quadratic SNAP lower; grey: within a factor of 1.26.



Figure 10: Nb: ratio of 10-draw mean held-out errors, quadratic SNAP over BLADE, per group and included fraction (as Fig. 9). Force cells are blank for the perfect crystals ("ground"), whose DFT forces vanish by symmetry.

Table 12: Mo: per-group mean held-out energy error (eV/atom) at selected fractions and full-data training error, quadratic SNAP / BLADE-qSNAP (10-draw means).

group	f=0.10	f=0.30	f=0.50	f=0.80	full data
BCC_elastic	0.0771 / 0.0677	0.0566 / 0.0702	0.0558 / 0.0697	0.0518 / 0.0679	0.0540 / 0.0716
BCC_interstitial	0.0497 / 0.0656	0.0464 / 0.0736	0.0577 / 0.0702	0.0526 / 0.0839	0.0480 / 0.0542
BCC_vacancy	0.0574 / 0.0481	0.0525 / 0.0488	0.0518 / 0.0522	0.0489 / 0.0468	0.0464 / 0.0684
DHCP_elastic	0.0734 / 0.0493	0.0520 / 0.0421	0.0446 / 0.0409	0.0404 / 0.0418	0.0380 / 0.0490
DHCP_vacancy	0.0342 / 0.0405	0.0268 / 0.0445	0.0240 / 0.0450	0.0226 / 0.0431	0.0216 / 0.0540
FCC_elastic	0.0425 / 0.0401	0.0368 / 0.0476	0.0354 / 0.0427	0.0313 / 0.0378	0.0311 / 0.0466
FCC_vacancy	0.0294 / 0.0395	0.0202 / 0.0464	0.0186 / 0.0447	0.0162 / 0.0468	0.0184 / 0.0555
SC_elastic	0.0739 / 0.0611	0.0584 / 0.0634	0.0571 / 0.0612	0.0479 / 0.0612	0.0531 / 0.0640
SC_vacancy	0.0731 / 0.0380	0.0568 / 0.0342	0.0736 / 0.0403	0.0729 / 0.0362	0.0731 / 0.0375
SH_elastic	0.123 / 0.101	0.0912 / 0.0910	0.0851 / 0.0812	0.0862 / 0.0864	0.0809 / 0.0895
SH_vacancy	0.0507 / 0.0397	0.0411 / 0.0451	0.0455 / 0.0395	0.0436 / 0.0456	0.0461 / 0.0853

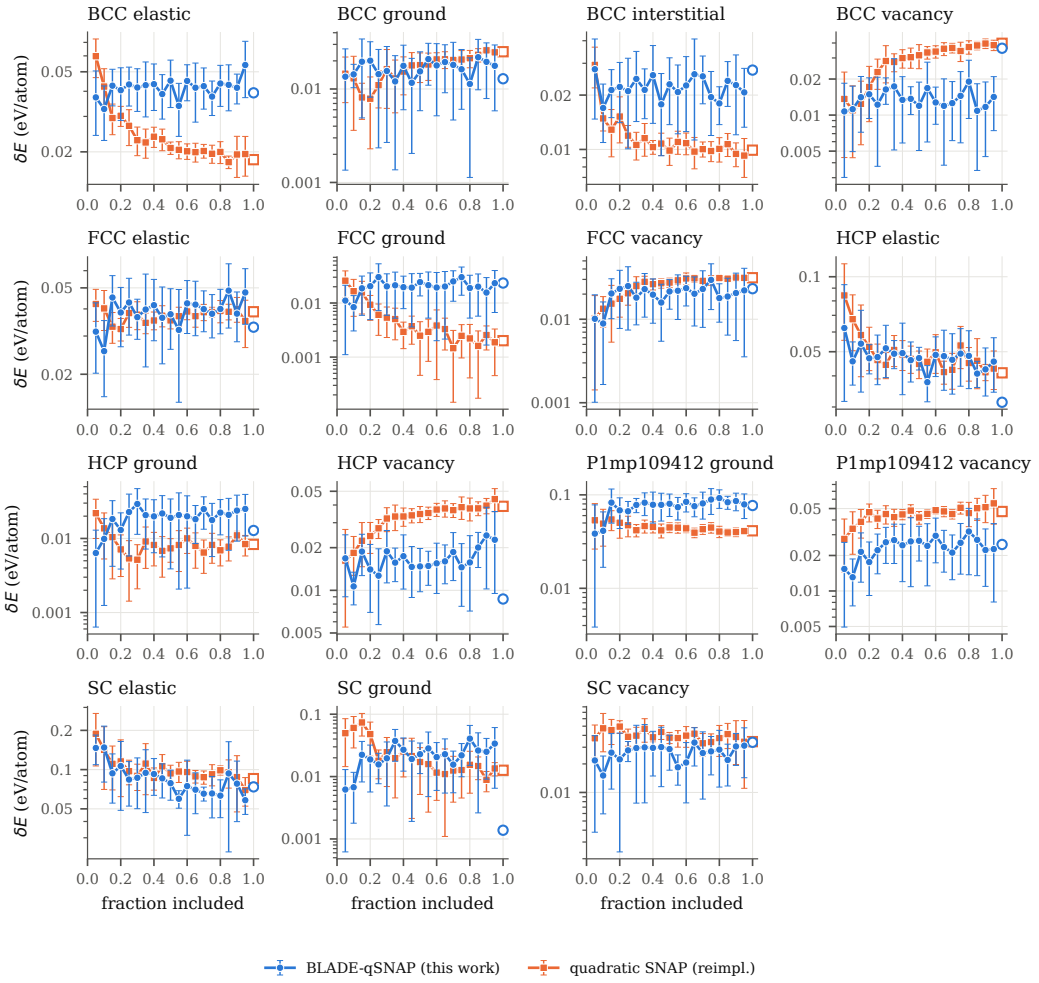


Figure 11: Nb: per-group held-out energy errors (mean ± std over 10 draws; open symbols: full-data training errors).

Table 13: Nb: per-group mean held-out energy error (eV/atom) at selected fractions and full-data training error, quadratic SNAP / BLADE-qSNAP (10-draw means).

group	f=0.10	f=0.30	f=0.50	f=0.80	full data
BCC_elastic	0.0422 / 0.0327	0.0229 / 0.0418	0.0209 / 0.0452	0.0200 / 0.0437	0.0183 / 0.0393
BCC_ground	0.0130 / 0.0143	0.0163 / 0.0156	0.0180 / 0.0154	0.0214 / 0.0113	0.0250 / 0.0129
BCC_interstitial	0.0149 / 0.0170	0.0106 / 0.0246	0.0099 / 0.0229	0.0101 / 0.0180	0.0099 / 0.0275
BCC_vacancy	0.0110 / 0.0112	0.0283 / 0.0164	0.0316 / 0.0120	0.0370 / 0.0190	0.0398 / 0.0362
FCC_elastic	0.0402 / 0.0256	0.0376 / 0.0367	0.0354 / 0.0378	0.0391 / 0.0400	0.0388 / 0.0330
FCC_ground	0.0165 / 0.0084	0.0054 / 0.0203	0.0025 / 0.0244	0.0022 / 0.0189	0.0020 / 0.0234
FCC_vacancy	0.0134 / 0.0089	0.0250 / 0.0181	0.0275 / 0.0215	0.0312 / 0.0179	0.0314 / 0.0232
HCP_elastic	0.0675 / 0.0457	0.0443 / 0.0516	0.0447 / 0.0472	0.0450 / 0.0481	0.0411 / 0.0313
HCP_ground	0.0137 / 0.0099	0.0052 / 0.0292	0.0074 / 0.0190	0.0069 / 0.0224	0.0083 / 0.0128
HCP_vacancy	0.0184 / 0.0107	0.0321 / 0.0189	0.0346 / 0.0148	0.0378 / 0.0157	0.0391 / 0.0087
P1mp109412_ground	0.0492 / 0.0412	0.0418 / 0.0780	0.0448 / 0.0803	0.0405 / 0.0924	0.0413 / 0.0769
P1mp109412_vacancy	0.0337 / 0.0131	0.0449 / 0.0259	0.0420 / 0.0267	0.0458 / 0.0320	0.0472 / 0.0248
SC_elastic	0.142 / 0.148	0.0896 / 0.0865	0.0935 / 0.0787	0.0988 / 0.0631	0.0853 / 0.0737
SC_ground	0.0603 / 0.0067	0.0246 / 0.0196	0.0171 / 0.0228	0.0154 / 0.0406	0.0126 / 0.0014
SC_vacancy	0.0475 / 0.0151	0.0397 / 0.0294	0.0379 / 0.0287	0.0374 / 0.0283	0.0344 / 0.0338

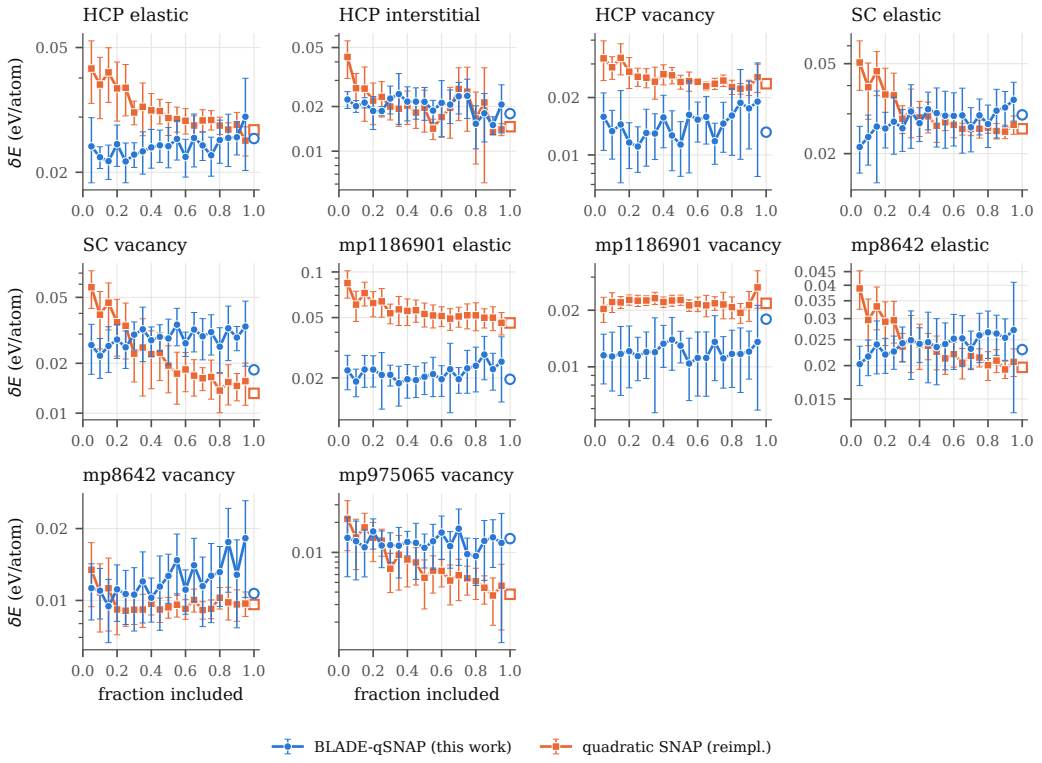


Figure 12: Re: per-group held-out energy errors (mean ± std over 10 draws; open symbols: full-data training errors).

Table 14: Re: per-group mean held-out energy error (eV/atom) at selected fractions and full-data training error, quadratic SNAP / BLADE-qSNAP (10-draw means).

group	f=0.10	f=0.30	f=0.50	f=0.80	full data
HCP_elastic	0.0381 / 0.0223	0.0310 / 0.0228	0.0298 / 0.0242	0.0282 / 0.0253	0.0273 / 0.0256
HCP_interstitial	0.0266 / 0.0201	0.0200 / 0.0226	0.0196 / 0.0215	0.0166 / 0.0153	0.0146 / 0.0179
HCP_vacancy	0.0288 / 0.0133	0.0254 / 0.0130	0.0241 / 0.0113	0.0227 / 0.0161	0.0236 / 0.0132
SC_elastic	0.0394 / 0.0237	0.0285 / 0.0259	0.0266 / 0.0302	0.0261 / 0.0270	0.0258 / 0.0297
SC_vacancy	0.0393 / 0.0222	0.0229 / 0.0298	0.0194 / 0.0282	0.0137 / 0.0252	0.0132 / 0.0183
mp1186901_elastic	0.0609 / 0.0189	0.0534 / 0.0210	0.0528 / 0.0203	0.0517 / 0.0241	0.0461 / 0.0196
mp1186901_vacancy	0.0221 / 0.0114	0.0225 / 0.0120	0.0226 / 0.0130	0.0208 / 0.0117	0.0218 / 0.0179
mp8642_elastic	0.0296 / 0.0216	0.0236 / 0.0243	0.0225 / 0.0235	0.0201 / 0.0266	0.0197 / 0.0230
mp8642_vacancy	0.0108 / 0.0110	0.0091 / 0.0106	0.0094 / 0.0127	0.0103 / 0.0131	0.0096 / 0.0107
mp975065_vacancy	0.0141 / 0.0130	0.0069 / 0.0118	0.0056 / 0.0111	0.0052 / 0.0092	0.0038 / 0.0137

D Reimplementing the published Ta baselines

Figure 13 repeats the cutoff scan of Fig. 3 of Ref. [4] on the provided Ta data with our pipeline: unweighted full-data fits at $J_{\max} = 4$ with the ZBL reference, every 0.1 Å between 4.0 and 7.0 Å (`logs/rcut_scan/ta_scan.jsonl`). The quadratic curve has the qualitative shape of the published one, with several local minima, and passes within 8 % of the Table SII value at the published cutoff. On the provided data, however, its global minimum on this grid is 0.0048 eV/atom at 6.6 Å, and it reaches 0.0048–0.0120 eV/atom everywhere above 4.5 Å. We nevertheless keep the published 5.594 Å, because the comparison is meant to be anchored to the published protocol.

Linear SNAP is not reproduced. Our unweighted linear fits stay between 0.122 and 0.172 eV/atom above 4.5 Å, four to six times the 0.03041 eV/atom of Table SI. We traced this to the relative weight of energy and force rows (`code/snap/ta_weighting_test.py`). With 55 coefficients the linear model cannot fit both, so its energy error depends strongly on that weight, whereas quadratic SNAP fits both and barely depends on it. At the published cutoffs, our per-atom-energy rows give 0.153 (linear) and 0.0082 (quadratic), against 0.03041 and 0.00898 in Tables SI and SII. Weighting the energy rows by 10 gives 0.0297 and 0.0008; weighting each by its atom count, which is equivalent to fitting total energies, gives 0.0367 and 0.0019. No single weighting reproduces both published values on the provided data. We therefore use one convention throughout, FitSNAP's unweighted per-atom rows, which reproduces the quadratic model that anchors every comparison. Linear SNAP enters our study only as a qualitative control.

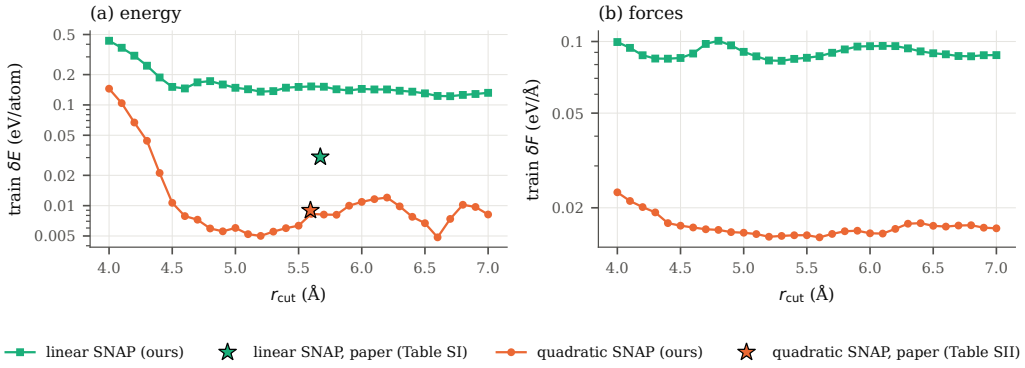


Figure 13: Ta, $J_{\max} = 4$: unweighted full-data training errors of our linear and quadratic SNAP versus cutoff (ZBL reference), with the published values of Tables SI and SII (stars).

E Cutoff and reference-potential selection for the transfer metals

Figure 14 shows the unweighted full-data training errors of quadratic SNAP over the scanned cutoffs and both reference choices (`logs/rcut_scan/`). For all three metals the lowest energy error is obtained without a reference potential. The ZBL repulsion, switched off between 4.0 and 4.8 Å, adds very large residuals for the strongly compressed frames of these datasets, and for Re its fits degrade by orders of magnitude at long cutoffs. As in the original paper, the energy error has several local minima in r_{cut}, while the force error grows smoothly with r_{cut}.

F Data pipeline

Ta. The FitSNAP JSON files are used unchanged. FitSNAP's scraper rotates every cell into the LAMMPS lower-triangular frame; the learned-descriptor pipeline reads these scraped structures, so both methods see identical coordinates and targets.

Mo, Nb, Re. The DeePMD arrays (`set.000/{coord,box,energy,force,virial}.npy`) of every system directory in the pre-staged archives are converted to one FitSNAP JSON file per frame

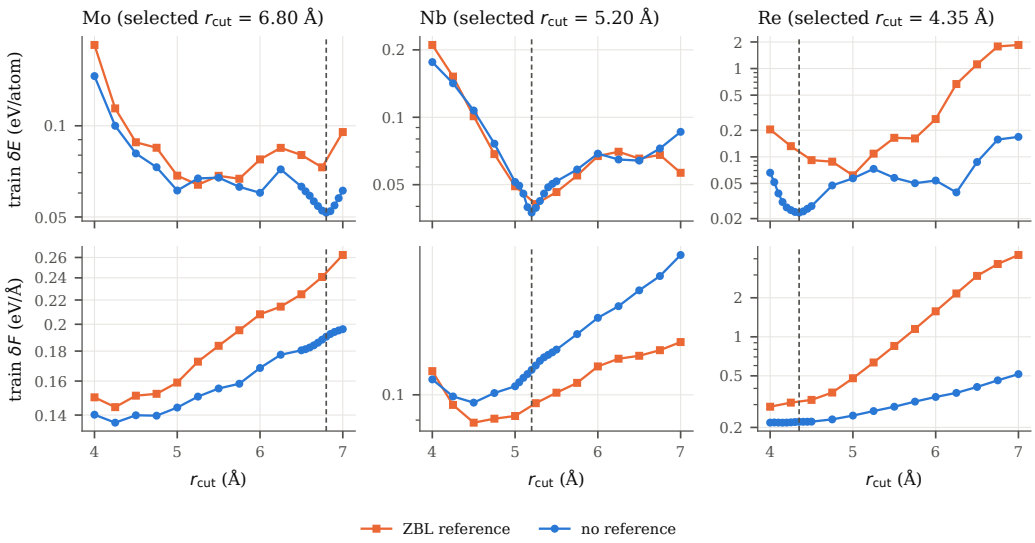


Figure 14: Quadratic-SNAP ($J_{\max} = 4$) full-data training errors versus cutoff for the transfer metals, with ZBL reference (squares) and without (circles). Dashed lines mark the selected cutoffs (no reference in all three cases).

by `code/data/deepmd_to_fitsnap.py`. Virials are stored as stresses but not fitted. Groups are named *parent structure_family*, with parent structures identified by `spglib`:

- Mo: BCC (mp-129), FCC (mp-8637), DHCP (mp-1066523), simple hexagonal (mp-1056004), simple cubic;
- Nb: BCC (mp-75), FCC (mp-8636), HCP, simple cubic, and a low-symmetry 12-atom phase (mp-109412);
- Re: HCP (mp-8), mp-1186901, mp-975065, mp-8642, simple cubic.

Exact duplicates (identical positions, cell and energy) are removed by `code/data/dedup.py`; the removed files are listed in `logs/data/`. Frames with identical geometry but different energies are kept.

G Code and log map

- `code/snap/`: FitSNAP drivers (design matrices, r_{cut} scans, SNAP-family CV sweeps, the Ta weighting test).
- `code/blade/`: descriptor (`core.py`, `sph.py`), bilevel training (`train.py`), sweeps (`run_sweep.py`), learned-projection control, inference and MD.
- `code/cv/`: split generation and metrics shared by all methods.
- `code/audit/`: invariance audit; `code/timing/`: MD timing and NVE energy conservation.
- `code/run_queue*.sh`, `code/run_final.sh`: the GPU work queue and the final audit/timing run, in the order in which they were executed.
- `code/analysis/`: every table, figure and number of this paper; `run_all_analysis.sh` regenerates all of them from the logs.
- `logs/cv/`: one JSON line per (material, method, draw), with seed, hyper-parameters, per-step history, timings and all error metrics on included and held-out data; `logs/cv_duplicates/`: same-seed re-runs (Sec. 4).
- `logs/audit/`, `logs/timing/`, `logs/rcut_scan/`, `logs/analysis/`, `logs/dev/`: per-sample audit drifts, timings, cutoff scans, derived analyses and the development record.
- `logs/preregistration.md`: protocol, frozen hyper-parameters and post-hoc additions, with times; `logs/paper_reference.md`: every value cited from Ref. [4].